

BIOLOGY

TENTH EDITION

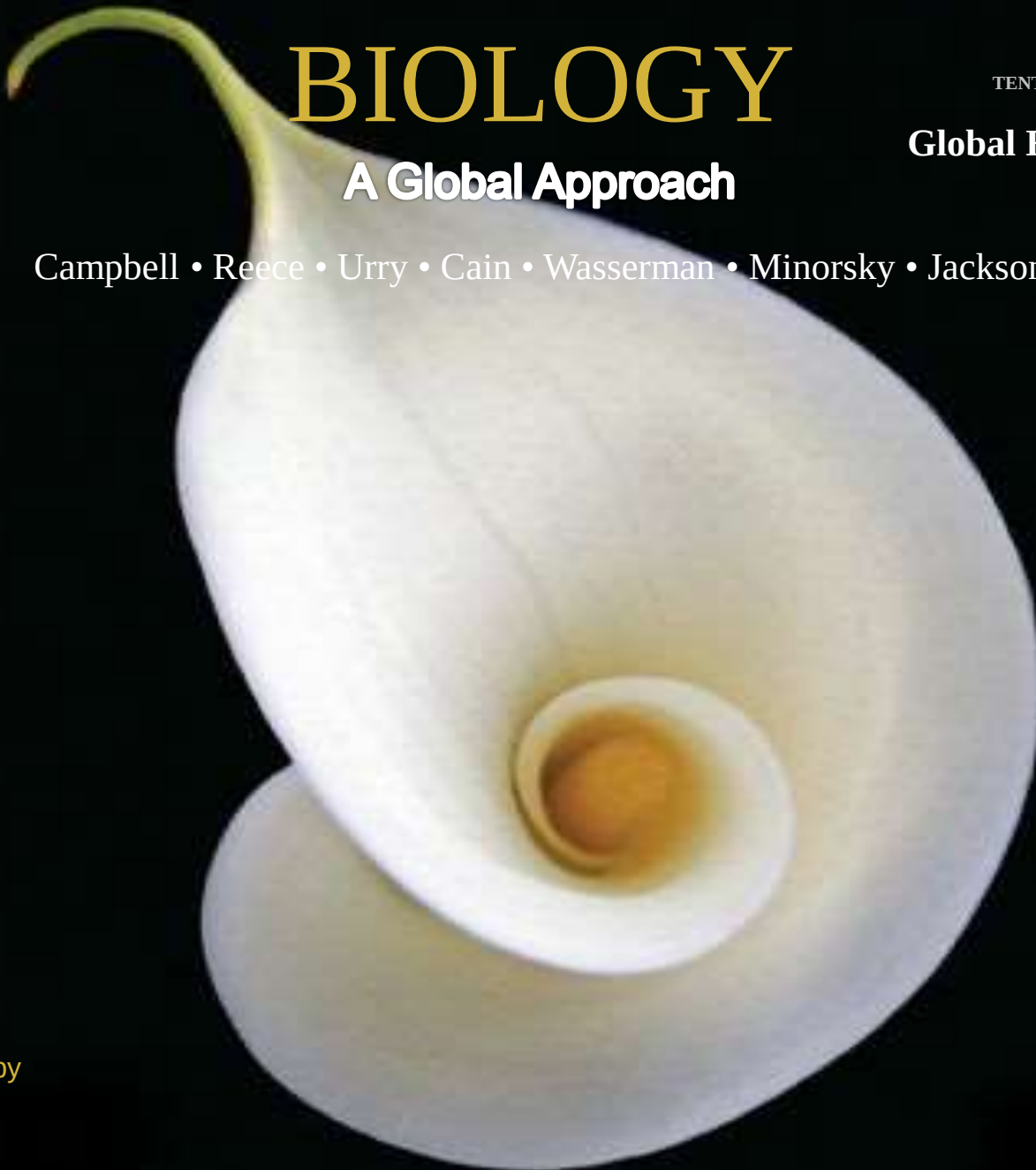
Global Edition

A Global Approach

Campbell • Reece • Urry • Cain • Wasserman • Minorsky • Jackson

Mitosis

12



Lecture Presentation by
Nicole Tunbridge and
Kathleen Fitzpatrick

The Key Roles of Cell Division

- a) The ability of organisms to produce more of their own kind best distinguishes living things from nonliving matter
- b) The continuity of life is based on the reproduction of cells, or **cell division**

Figure 12.1

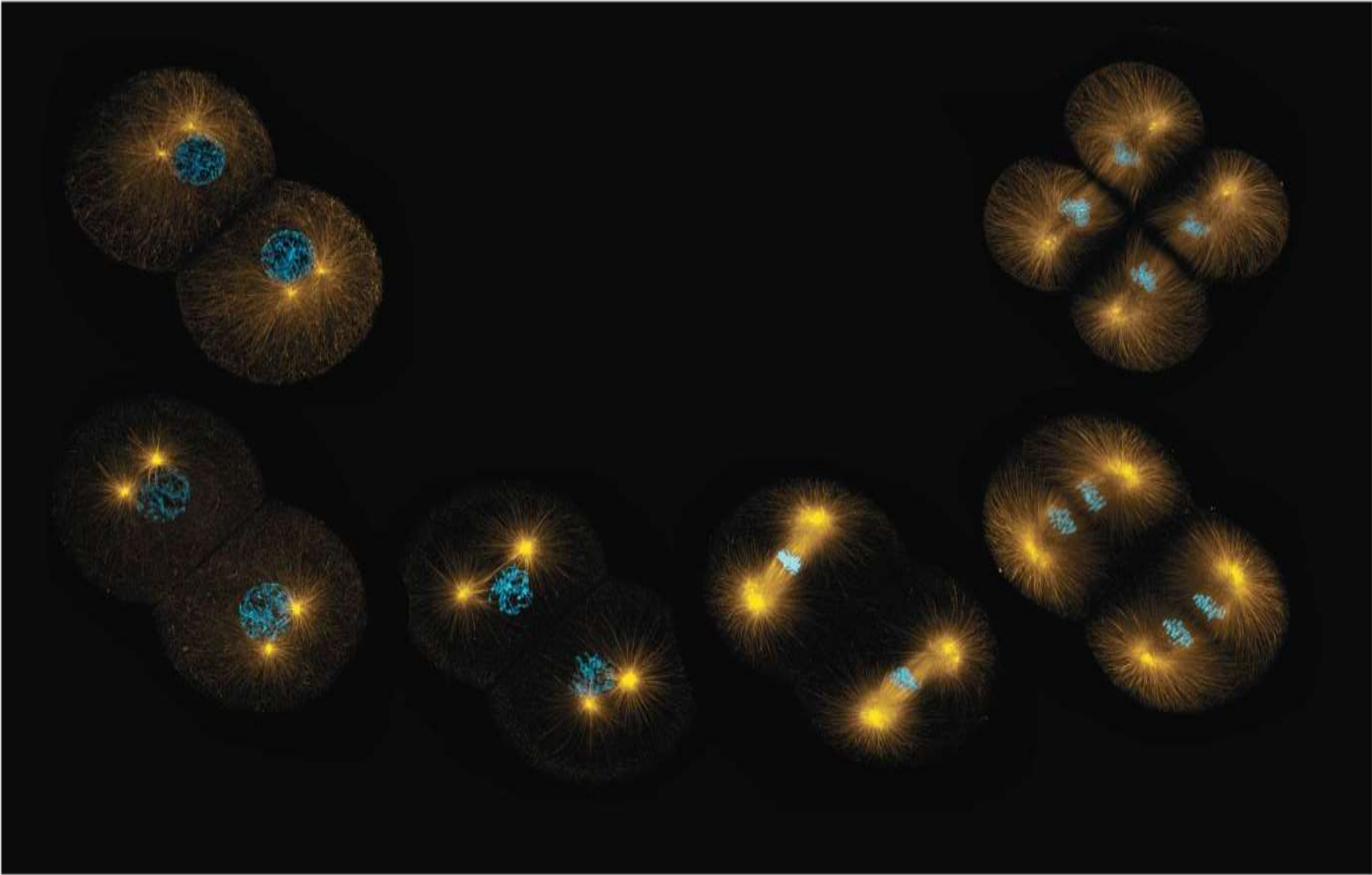
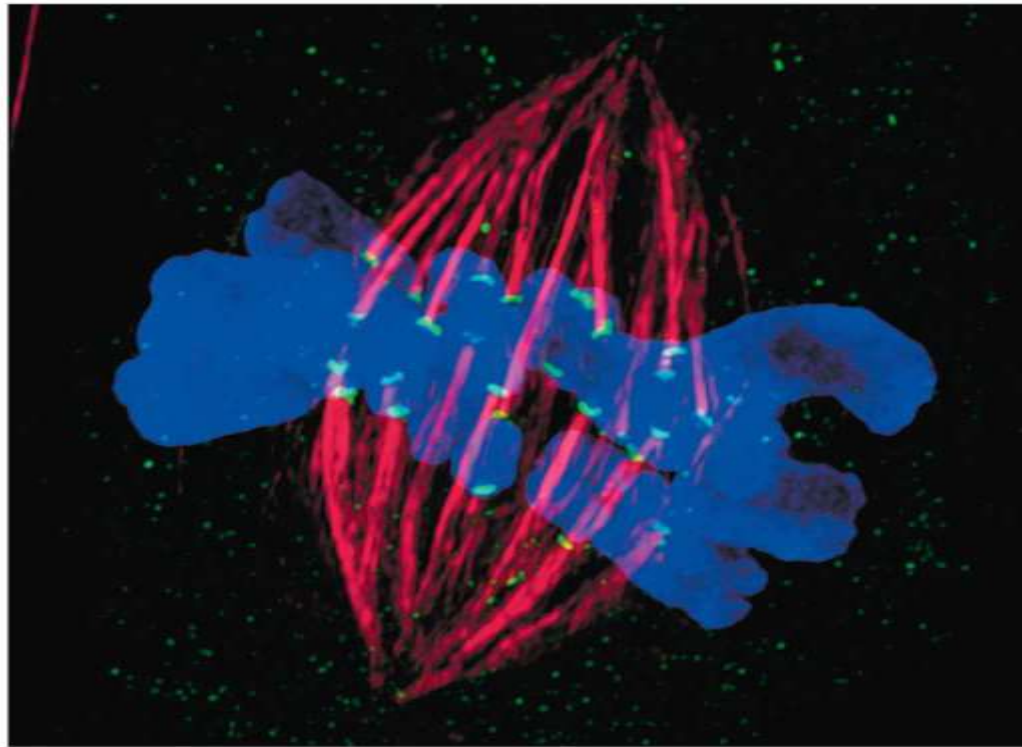
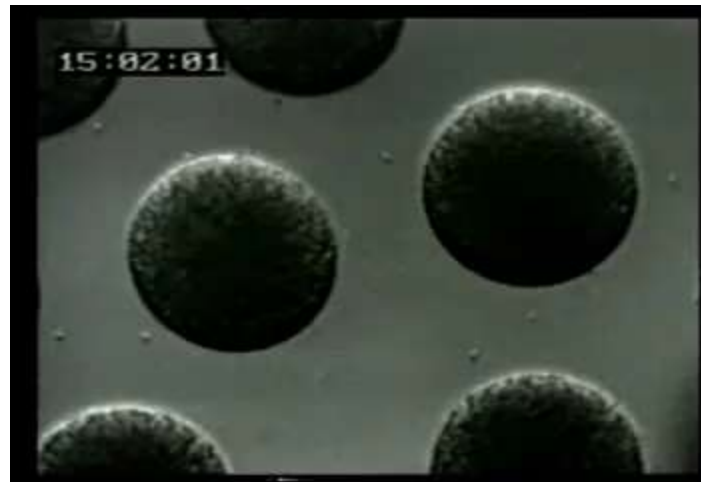


Figure 12.1a



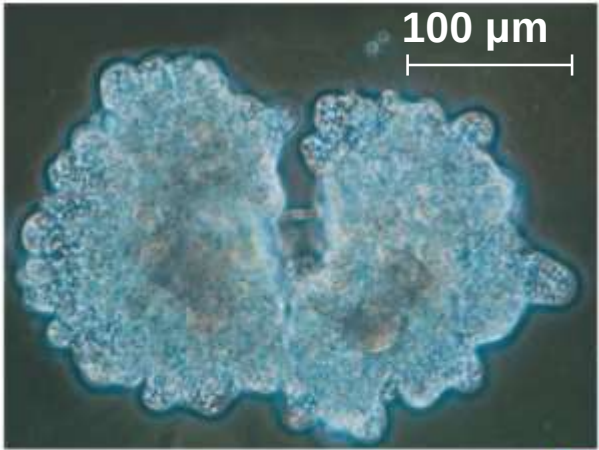
Chromosomes (blue) are moved by cell machinery (red) during division of a rat kangaroo cell.

Video: Sea Urchin Embryonic Development (time-lapse)



- a) In unicellular organisms, division of one cell reproduces the entire organism
- b) Multicellular eukaryotes depend on cell division for
 - a) Development from a fertilized cell
 - b) Growth
 - c) Repair
- c) Cell division is an integral part of the **cell cycle**, the life of a cell from formation to its own division

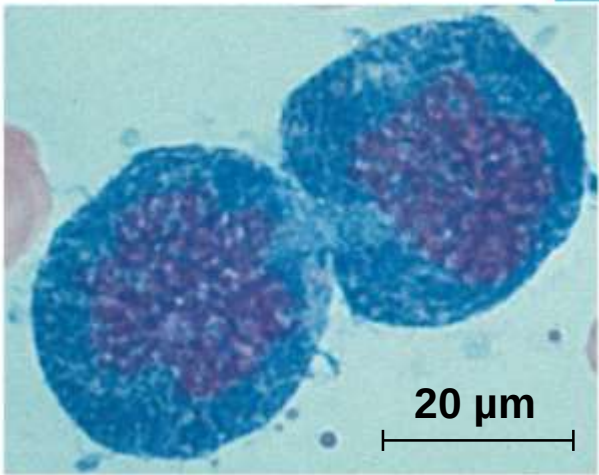
Figure 12.2



(a) Reproduction



(b) Growth and development



(c) Tissue renewal

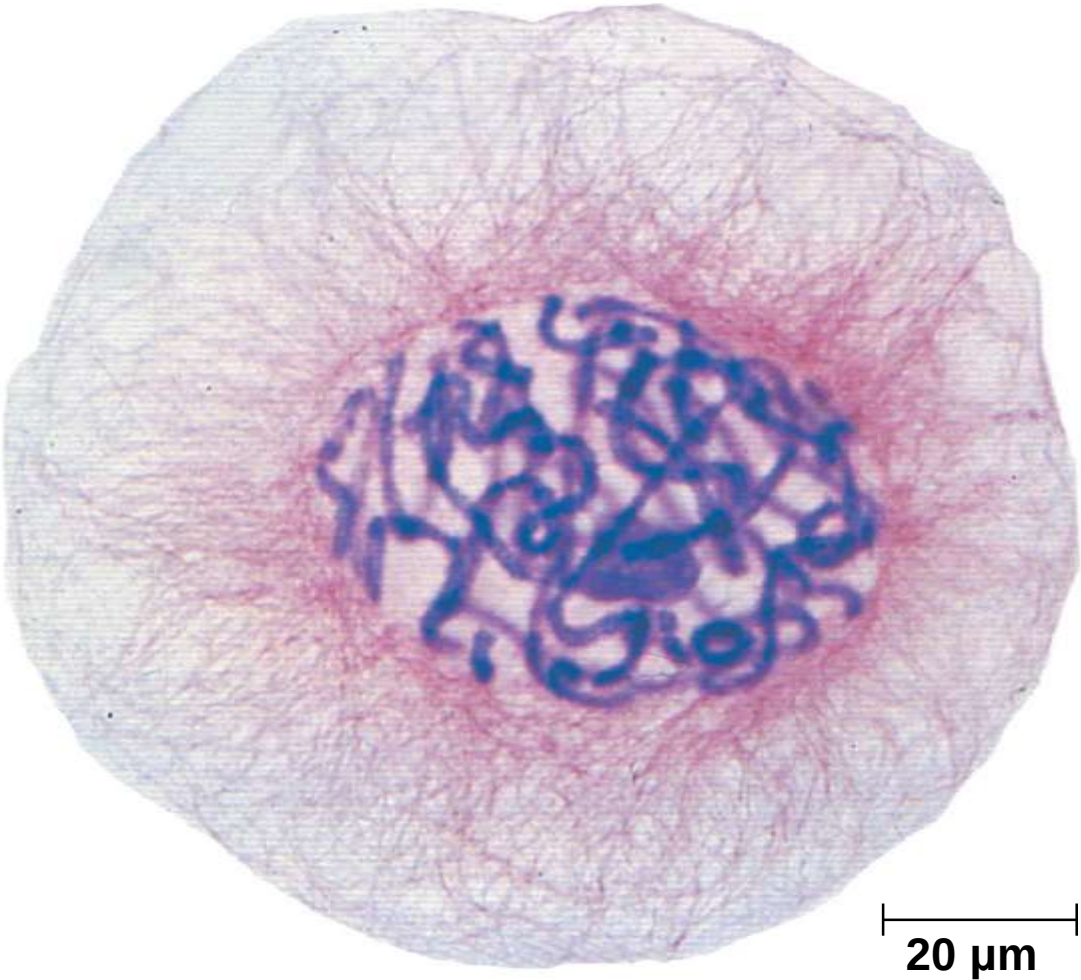
Concept 12.1: Most cell division results in genetically identical daughter cells

- a) Most cell division results in daughter cells with identical genetic information, DNA
- b) The exception is meiosis, a special type of division that can produce sperm and egg cells

Cellular Organization of the Genetic Material

- a) All the DNA in a cell constitutes the cell's **genome**
- b) A genome can consist of a single DNA molecule (common in prokaryotic cells) or a number of DNA molecules (common in eukaryotic cells)
- c) DNA molecules in a cell are packaged into **chromosomes**

Figure 12.3

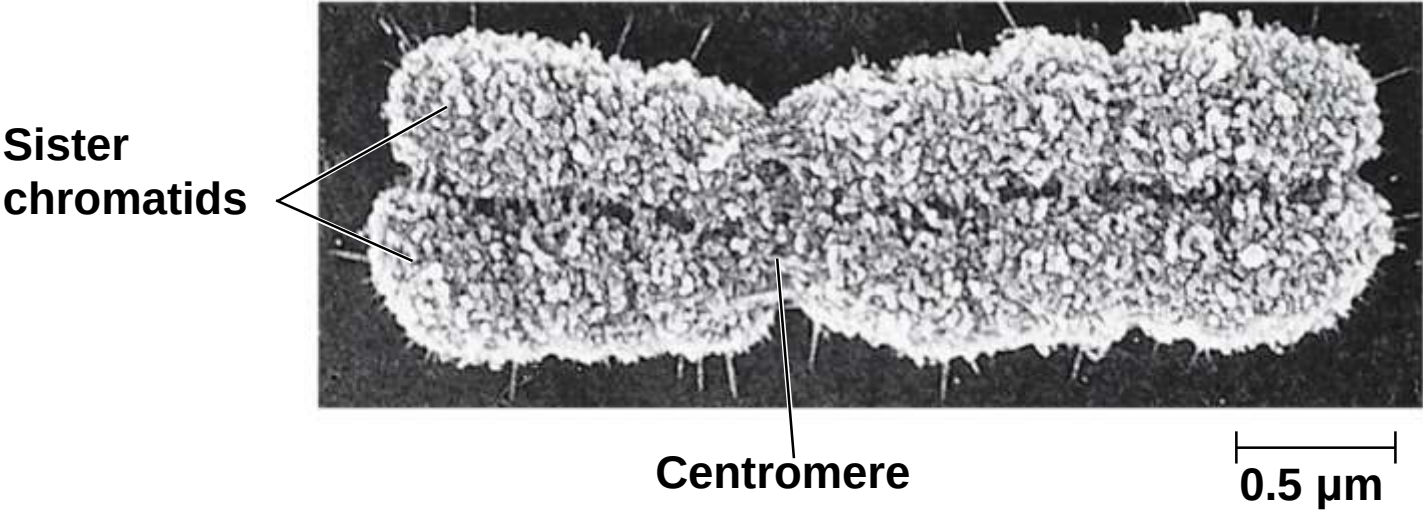


- a) Eukaryotic chromosomes consist of **chromatin**, a complex of DNA and protein that condenses during cell division
- b) Every eukaryotic species has a characteristic number of chromosomes in each cell nucleus
- c) **Somatic cells** (nonreproductive cells) have two sets of chromosomes
- d) **Gametes** (reproductive cells: sperm and eggs) have half as many chromosomes as somatic cells

Distribution of Chromosomes During Eukaryotic Cell Division

- a) In preparation for cell division, DNA is replicated and the chromosomes condense
- b) Each duplicated chromosome has two **sister chromatids** (joined copies of the original chromosome), attached along their lengths by cohesins
- c) The **centromere** is the narrow “waist” of the duplicated chromosome, where the two chromatids are most closely attached

Figure 12.4



- a) During cell division, the two sister chromatids of each duplicated chromosome separate and move into two nuclei
- b) Once separate, the chromatids are called chromosomes

Figure 12.5-1

1

Chromosomes

**Chromosomal
DNA molecules**

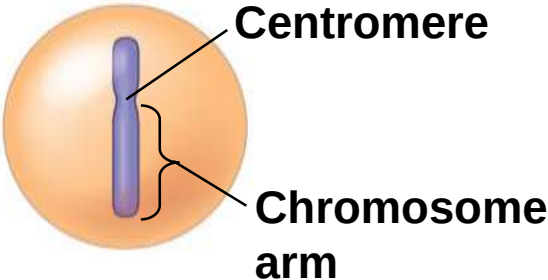


Figure 12.5-2

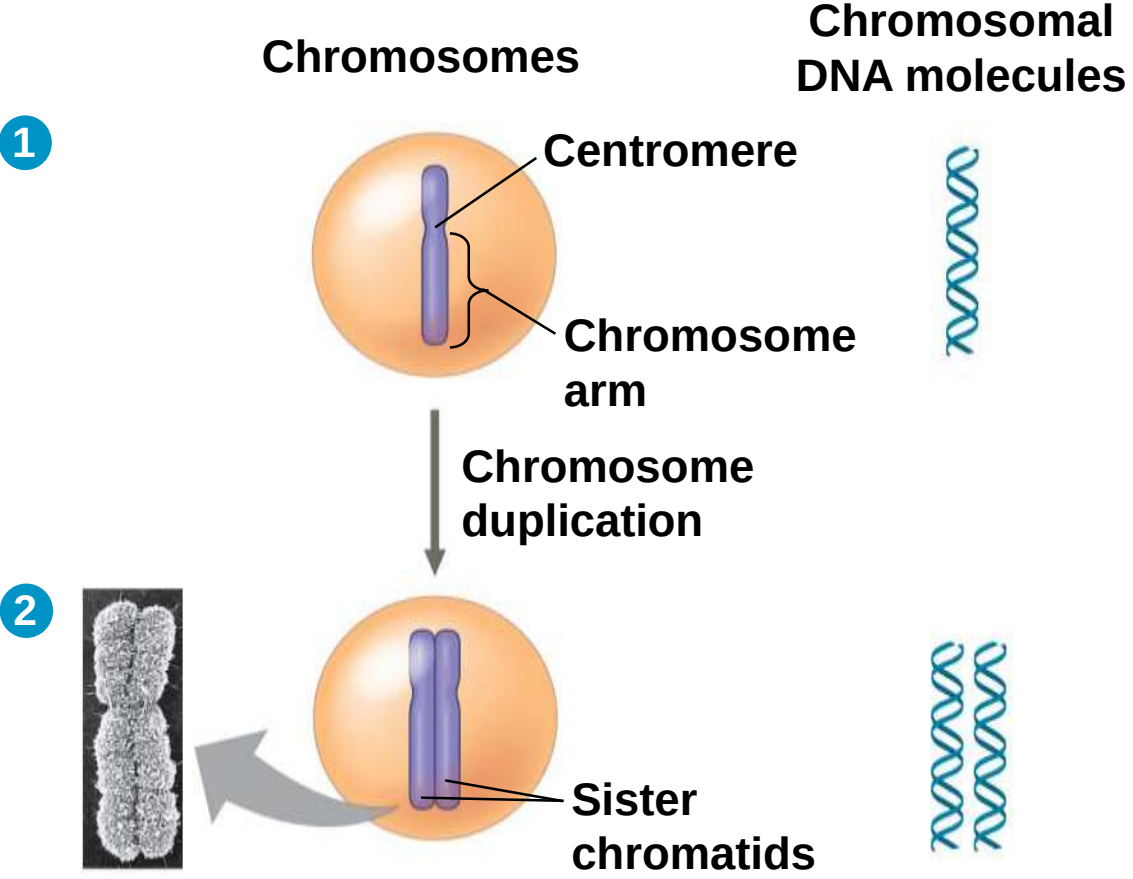
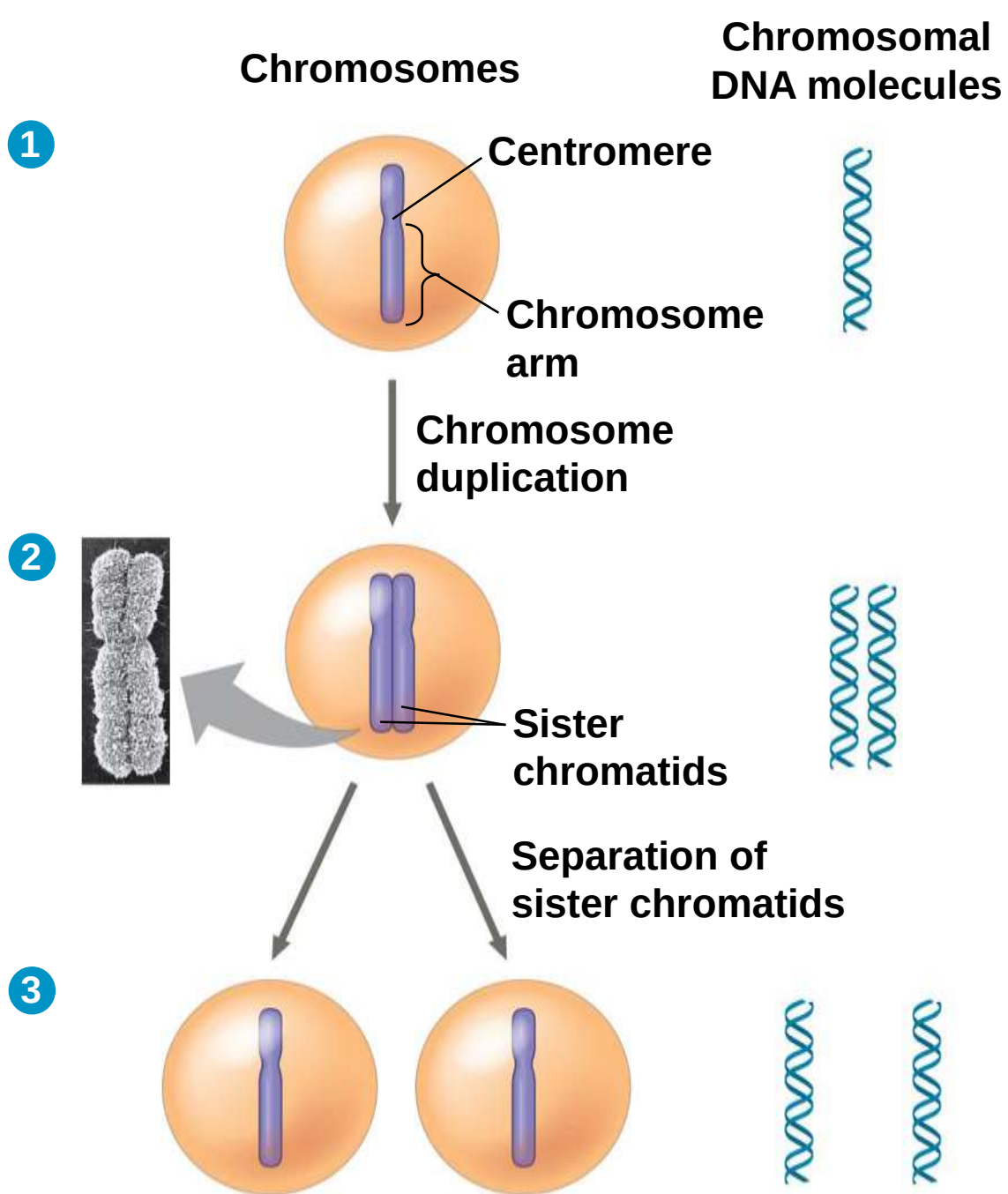


Figure 12.5-3



a) Eukaryotic cell division consists of

a)Mitosis, the division of the genetic material in the nucleus

b)Cytokinesis, the division of the cytoplasm

b) Gametes are produced by a variation of cell division called meiosis

c) Meiosis yields nonidentical daughter cells that have half as many chromosomes as the parent cell

Concept 12.2: The mitotic phase alternates with interphase in the cell cycle

- a) In 1882, the German anatomist Walther Flemming developed dyes to observe chromosomes during mitosis and cytokinesis

Phases of the Cell Cycle

a) The cell cycle consists of

a) Mitotic (M) phase (mitosis and cytokinesis)

b) Interphase (cell growth and copying of chromosomes in preparation for cell division)

a) Interphase (about 90% of the cell cycle) can be divided into subphases

a)G₁ phase (“first gap”)

b)S phase (“synthesis”)

c)G₂ phase (“second gap”)

b) The cell grows during all three phases, but chromosomes are duplicated only during the S phase

Figure 12.6

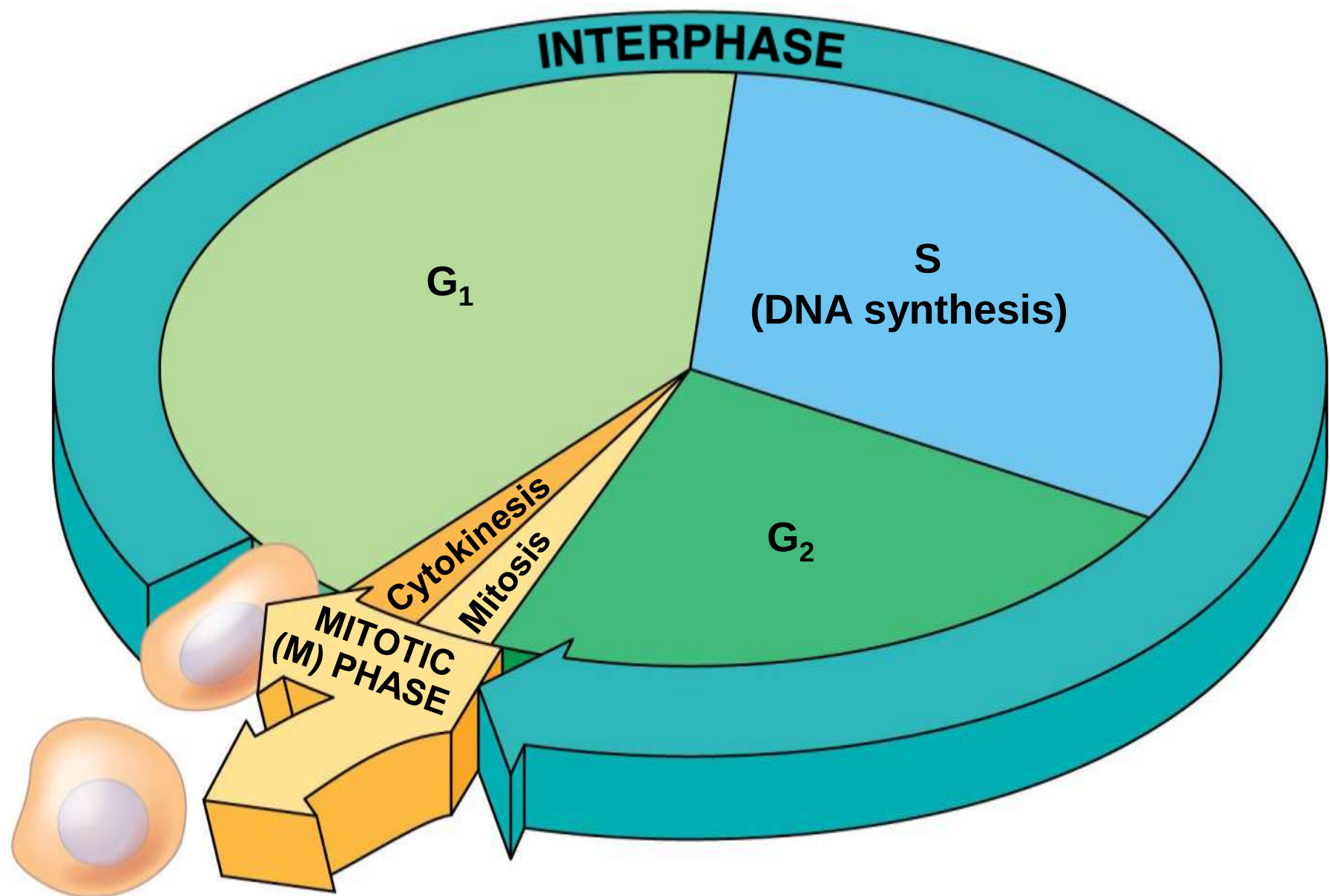
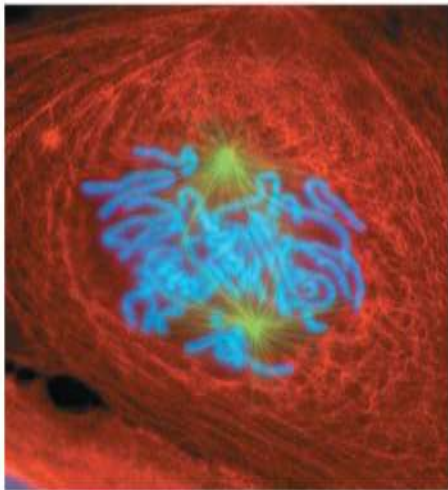
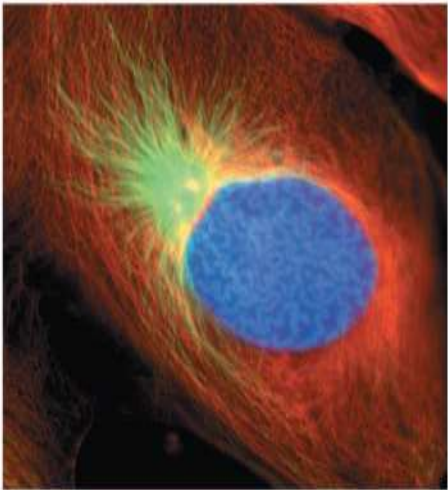
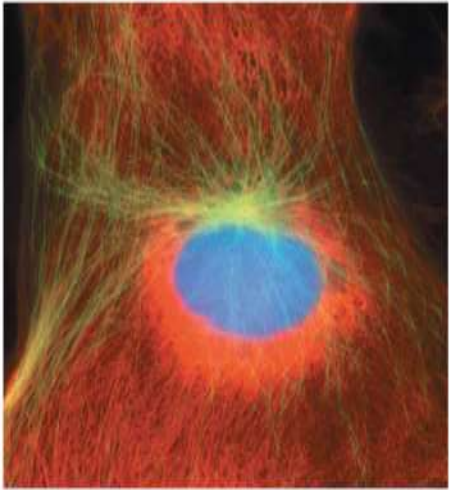


Figure 12.7a



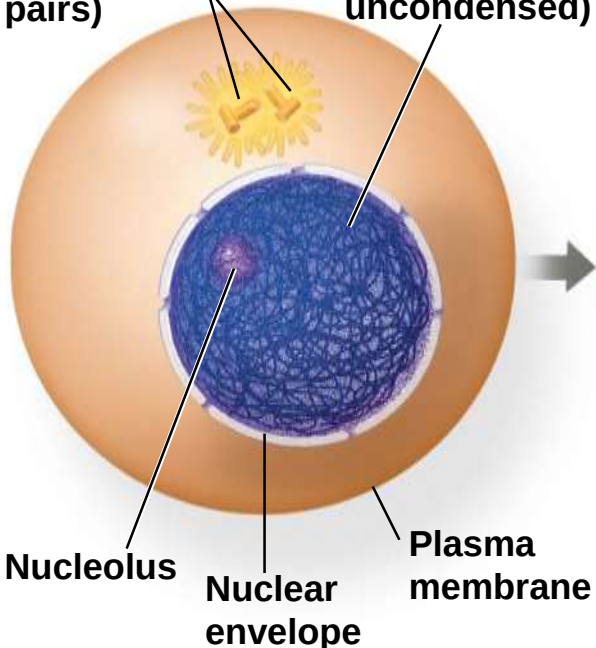
10 μm

G₂ of Interphase

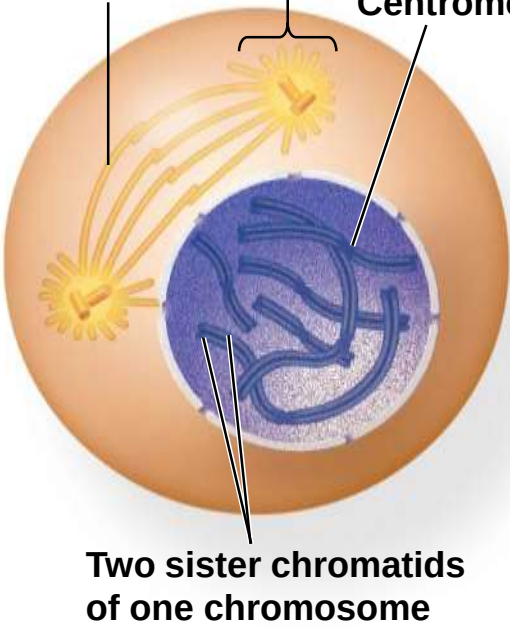
Prophase

Prometaphase

Centrosomes (with centriole pairs)
Chromosomes (duplicated, uncondensed)



Early mitotic spindle
Aster
Centromere



Fragments of nuclear envelope
Nonkinetochore microtubules

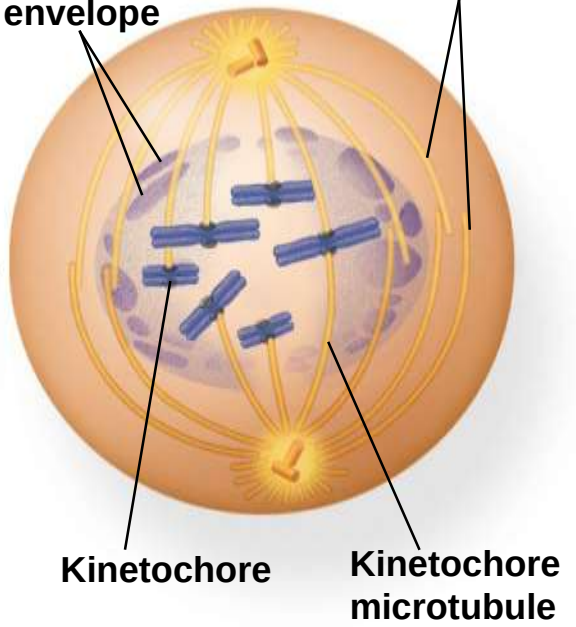
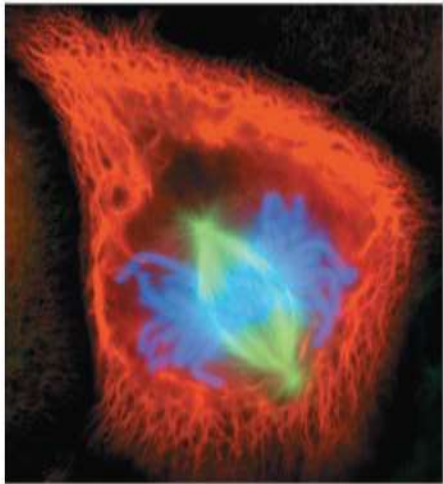
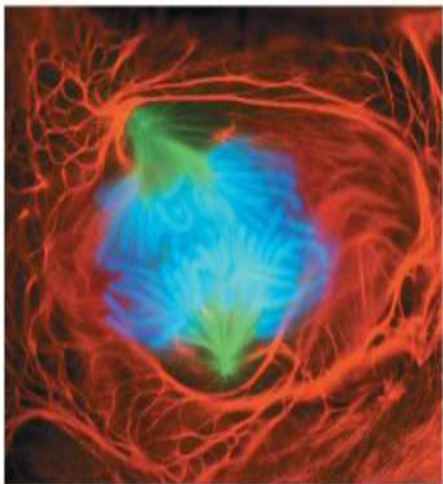


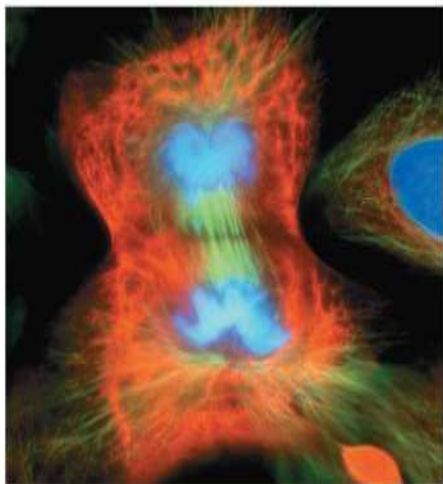
Figure 12.7b



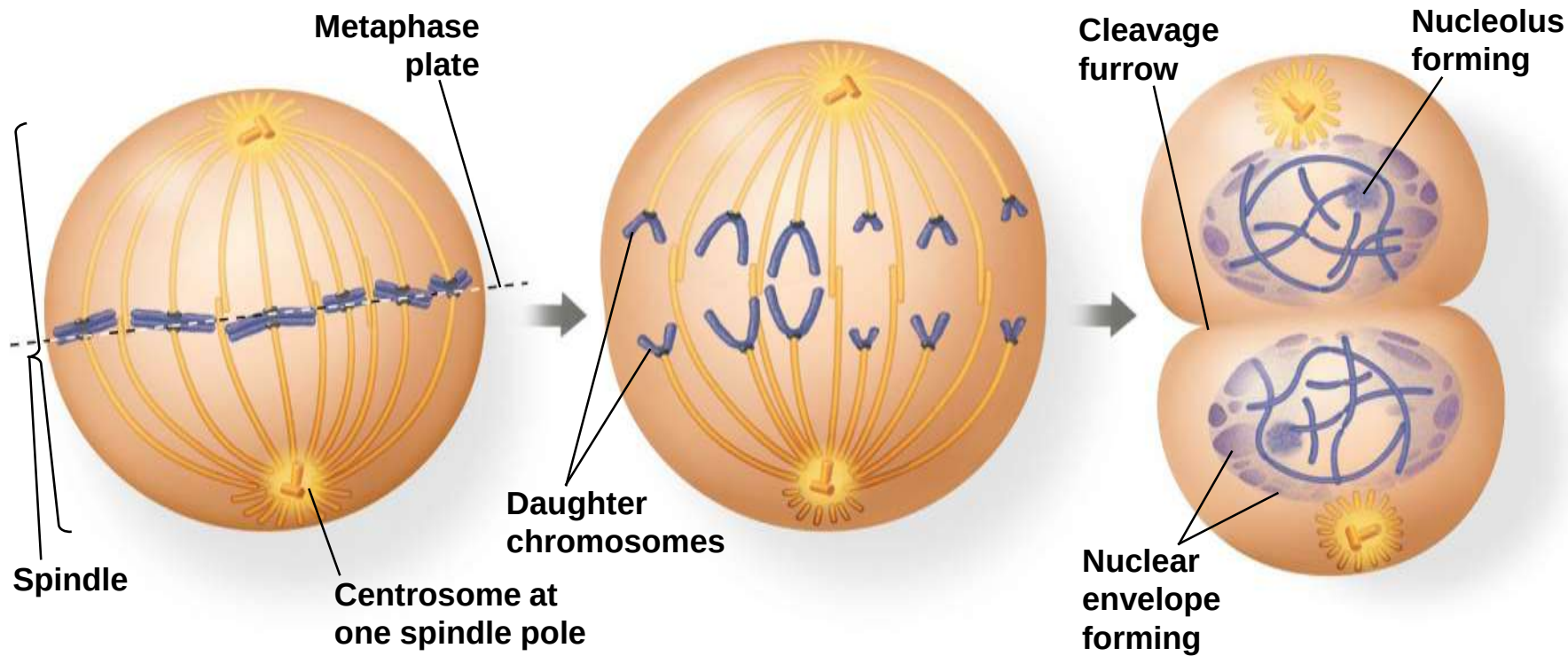
Metaphase



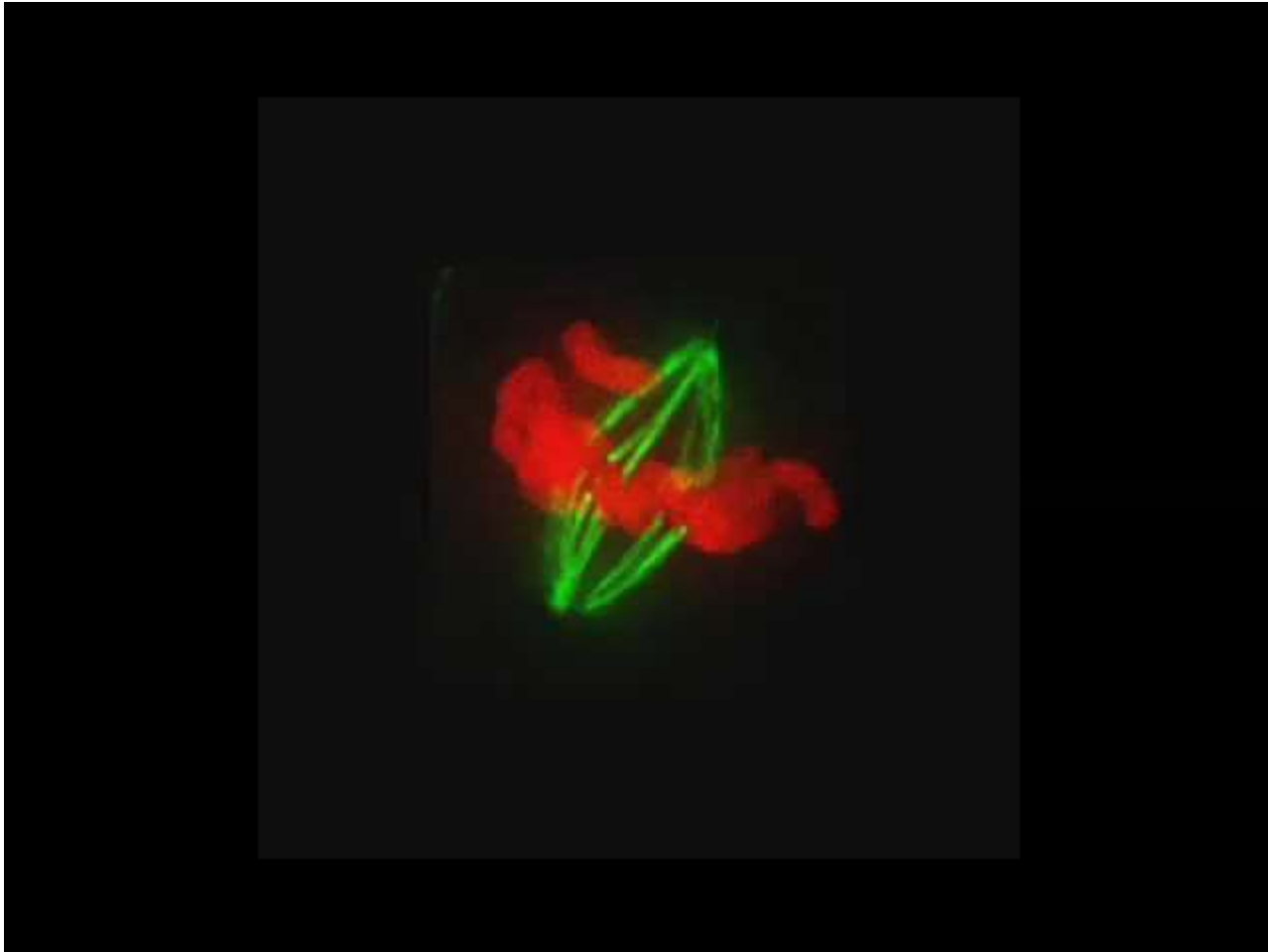
Anaphase



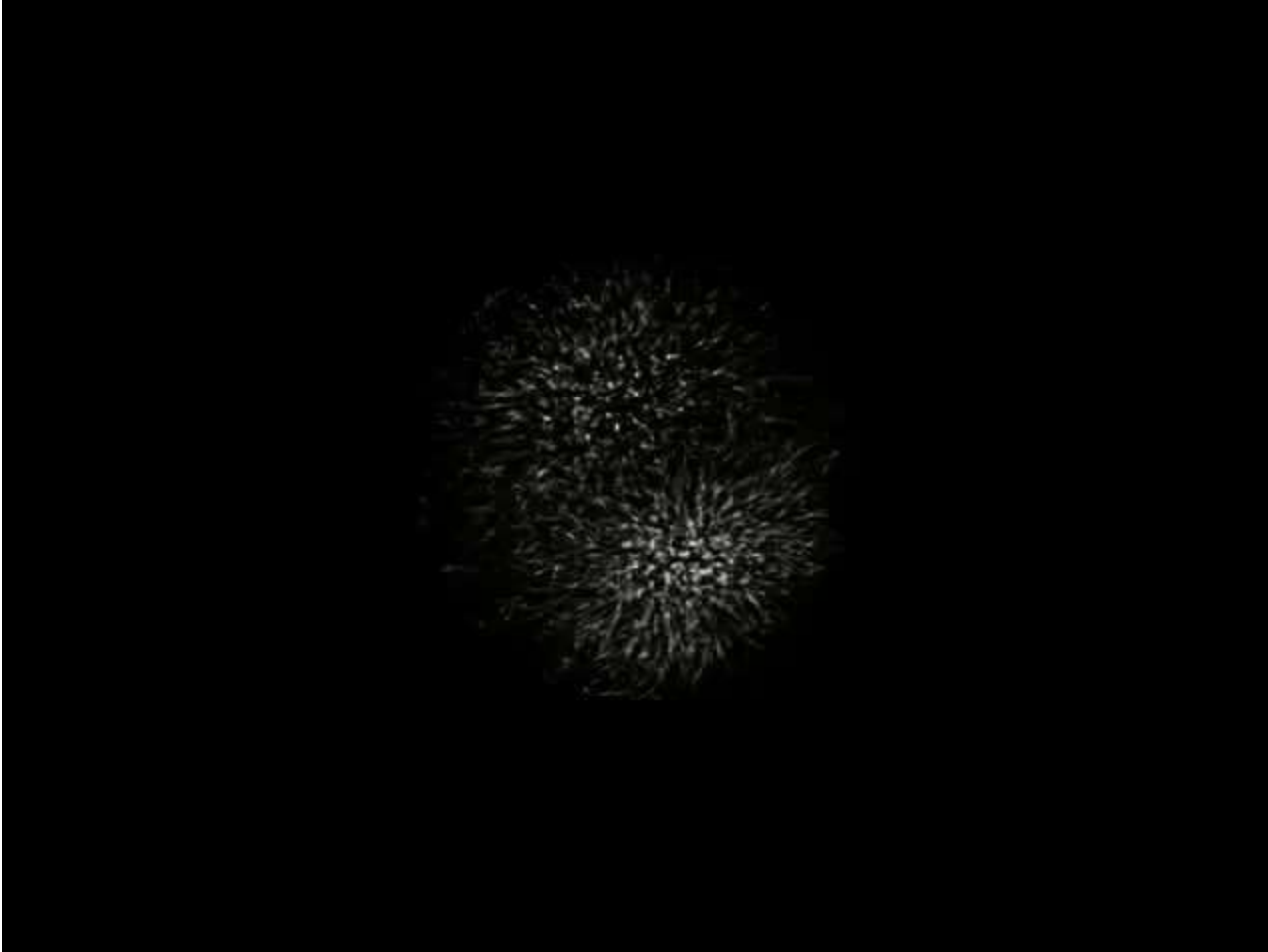
Telophase and Cytokinesis



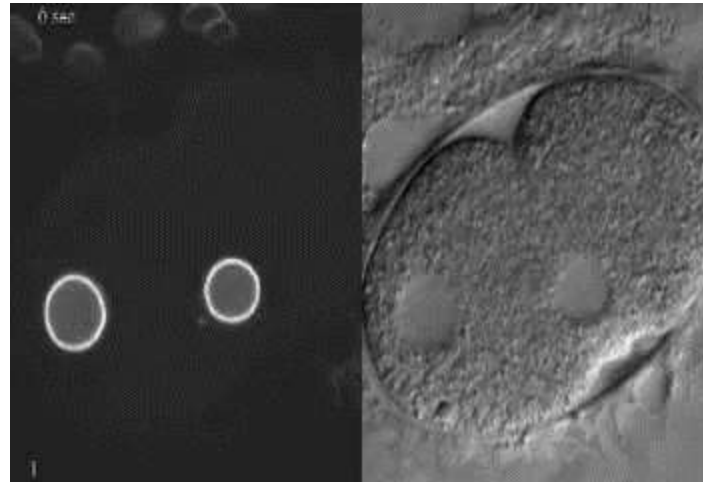
Video: Microtubules in Cell Division



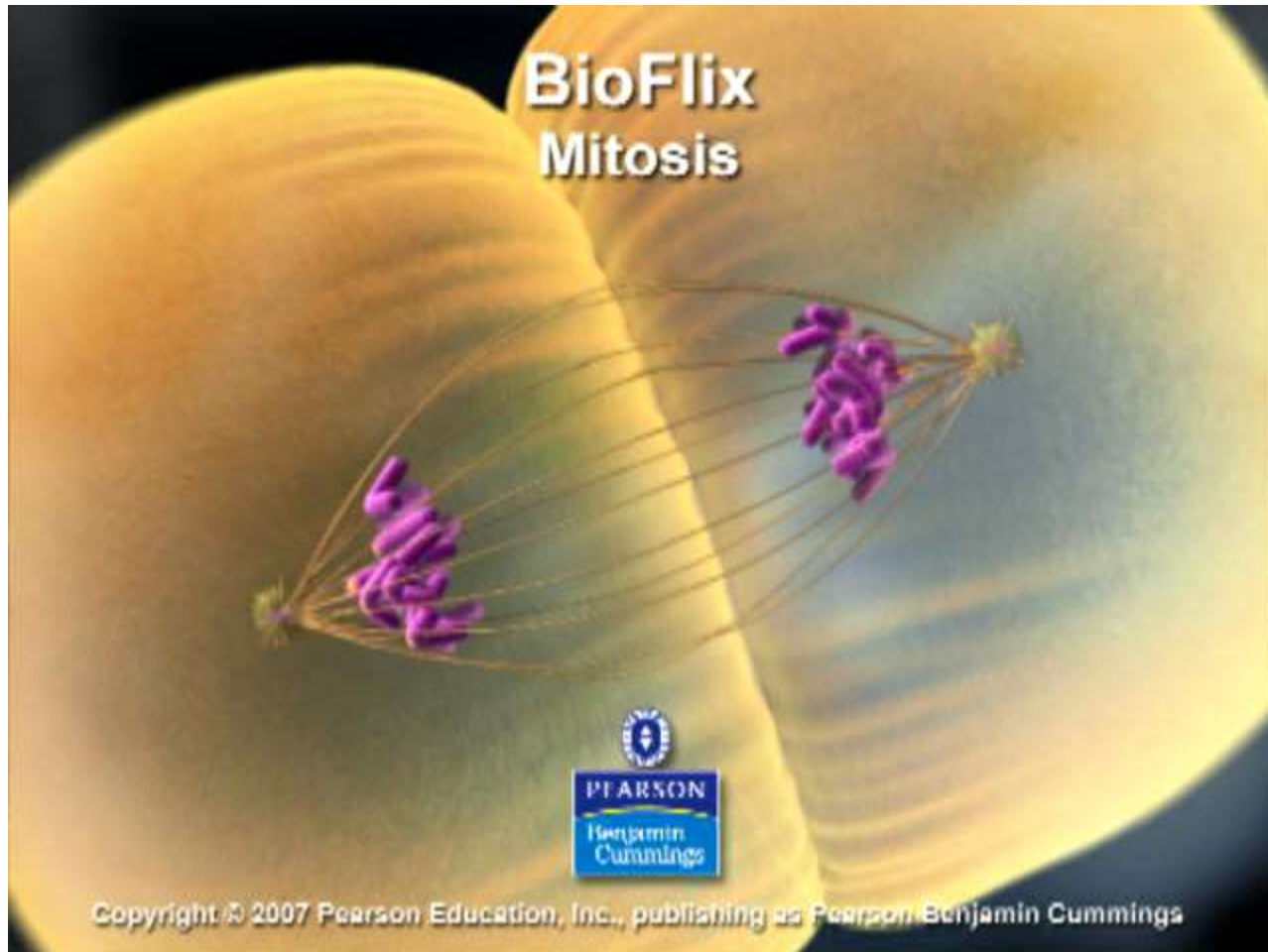
Video: Microtubules in Anaphase



Video: Nuclear Envelope Formation



BioFlix: Mitosis



Video: Animal Mitosis (time-lapse)



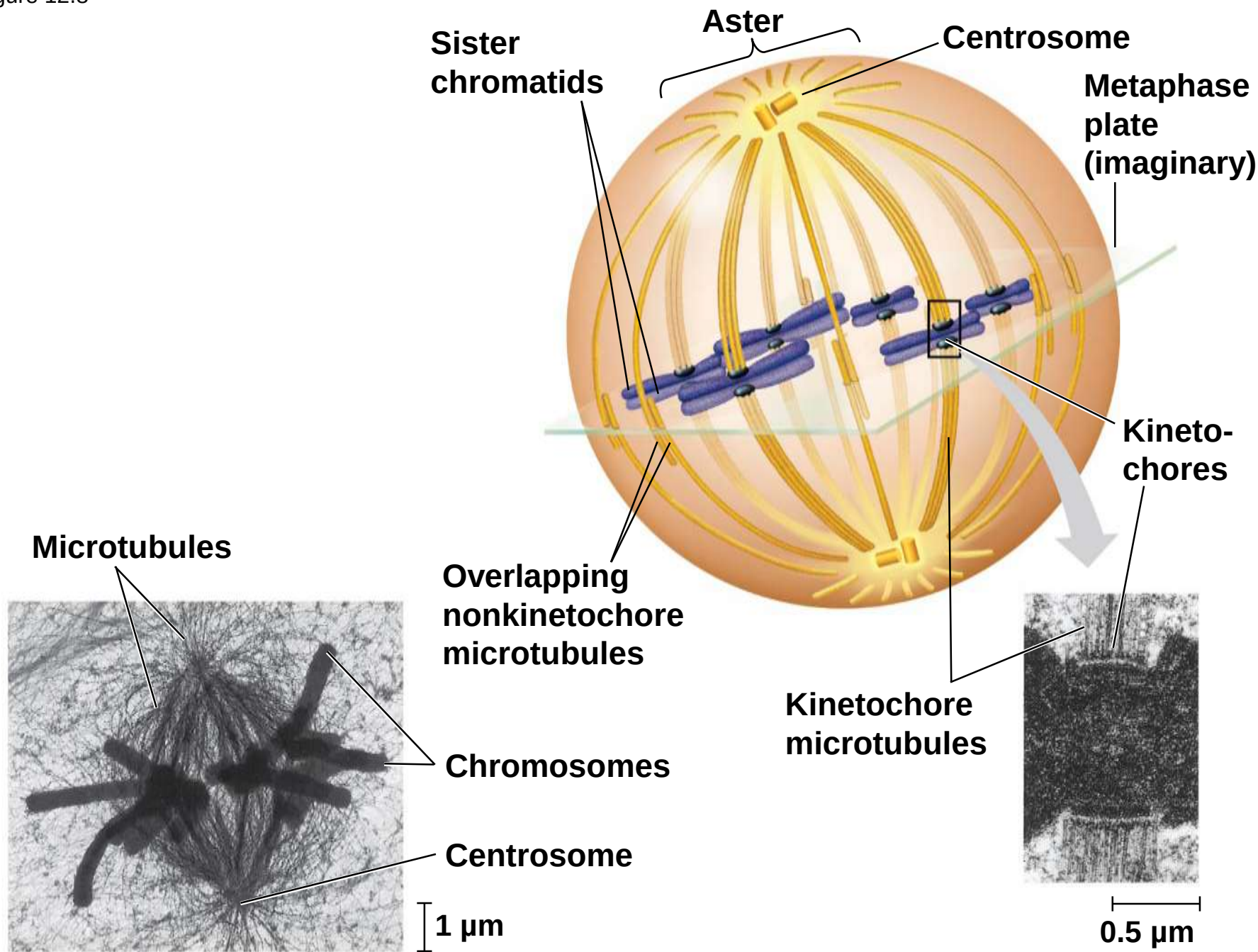
The Mitotic Spindle: *A Closer Look*

- a) The **mitotic spindle** is a structure made of microtubules that controls chromosome movement during mitosis
- b) In animal cells, assembly of spindle microtubules begins in the **centrosome**, the microtubule organizing center
- c) The centrosome replicates during interphase, forming two centrosomes that migrate to opposite ends of the cell during prophase and prometaphase

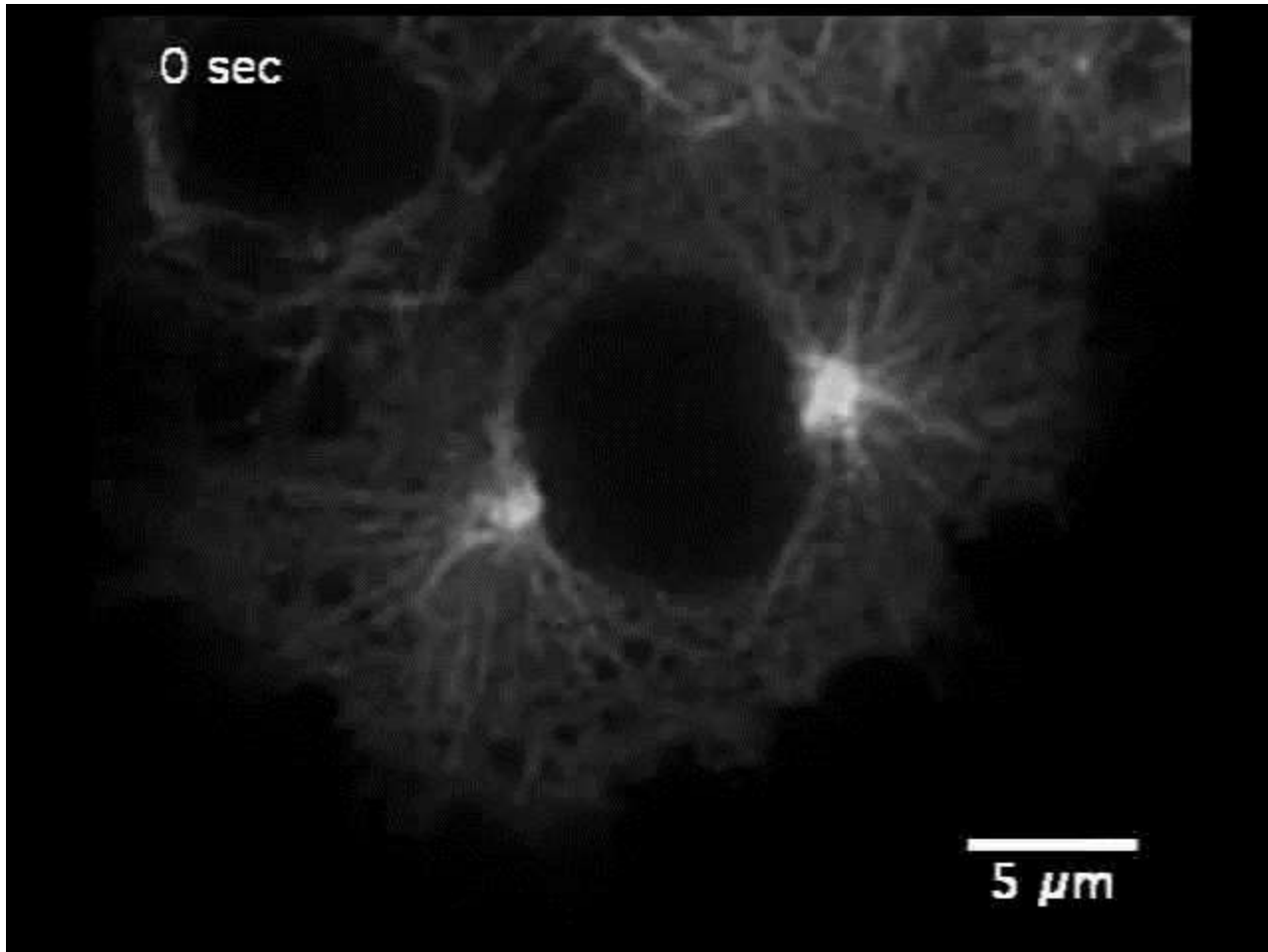
- a) An **aster** (a radial array of short microtubules) extends from each centrosome
- b) The spindle includes the centrosomes, the spindle microtubules, and the asters

- a) During prometaphase, some spindle microtubules attach to the kinetochores of chromosomes and begin to move the chromosomes
- b) **Kinetochores** are protein complexes associated with centromeres
- c) At metaphase, the chromosomes are all lined up at the **metaphase plate**, a plane midway between the spindle's two poles

Figure 12.8



Video: Spindle Formation During Mitosis



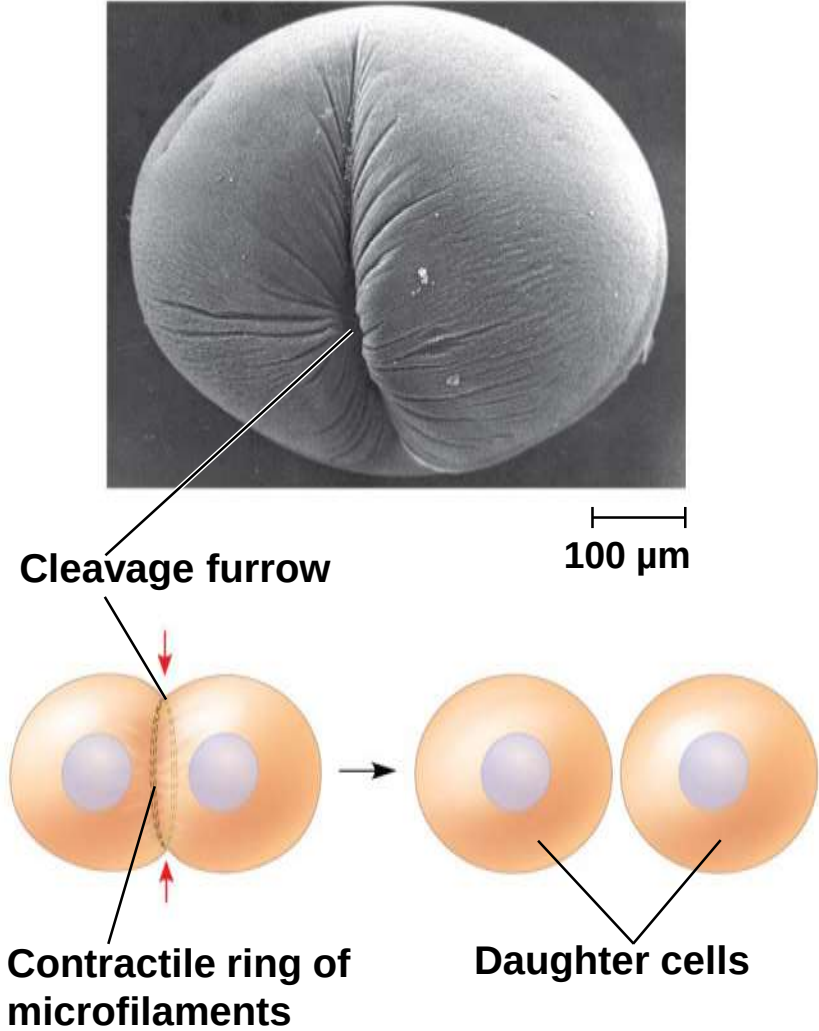
- a) In anaphase the cohesins are cleaved by an enzyme called separase
- b) Sister chromatids separate and move along the kinetochore microtubules toward opposite ends of the cell
- c) The microtubules shorten by depolymerizing at their kinetochore ends

- a) Nonkinetochore microtubules from opposite poles overlap and push against each other, elongating the cell
- b) In telophase, genetically identical daughter nuclei form at opposite ends of the cell
- c) Cytokinesis begins during anaphase or telophase and the spindle eventually disassembles

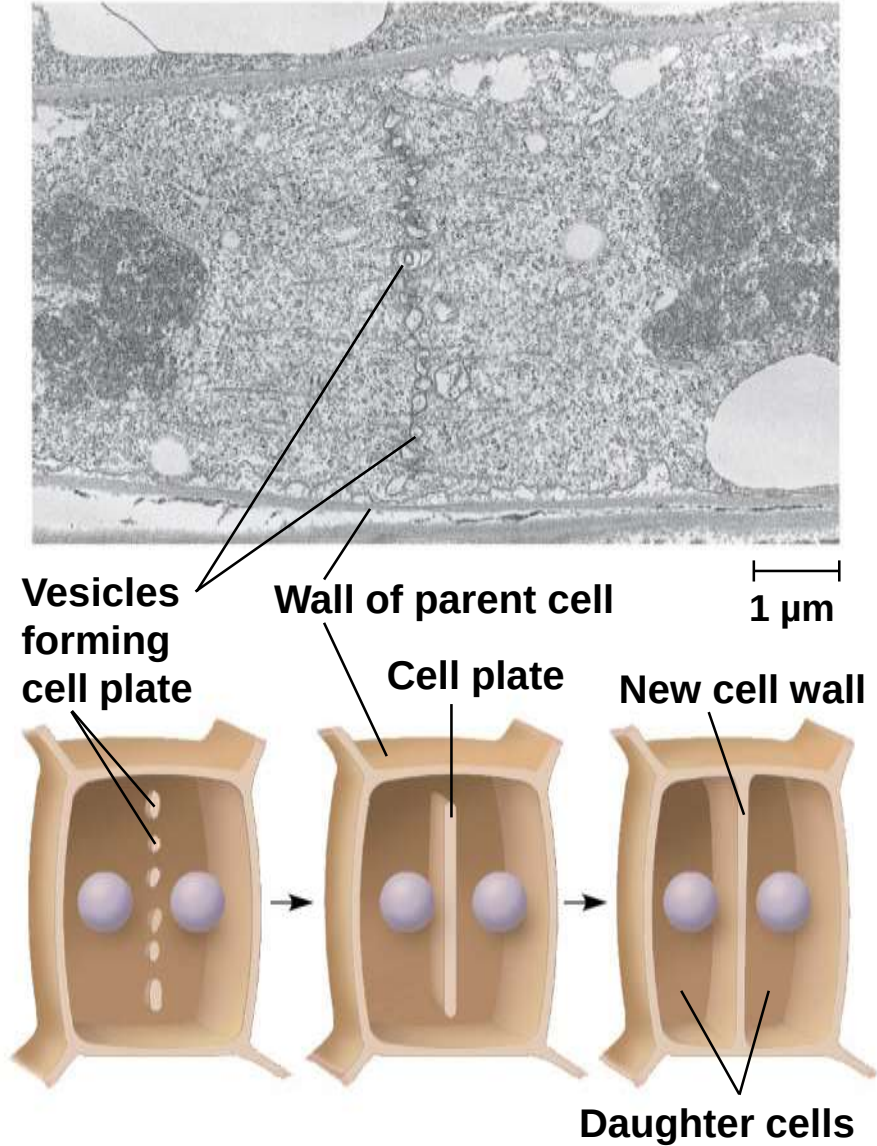
Cytokinesis: *A Closer Look*

- a) In animal cells, cytokinesis occurs by a process known as **cleavage**, forming a **cleavage furrow**
- b) In plant cells, a **cell plate** forms during cytokinesis

(a) Cleavage of an animal cell (SEM)



(b) Cell plate formation in a plant cell (TEM)



(a) Cleavage of an animal cell (SEM)

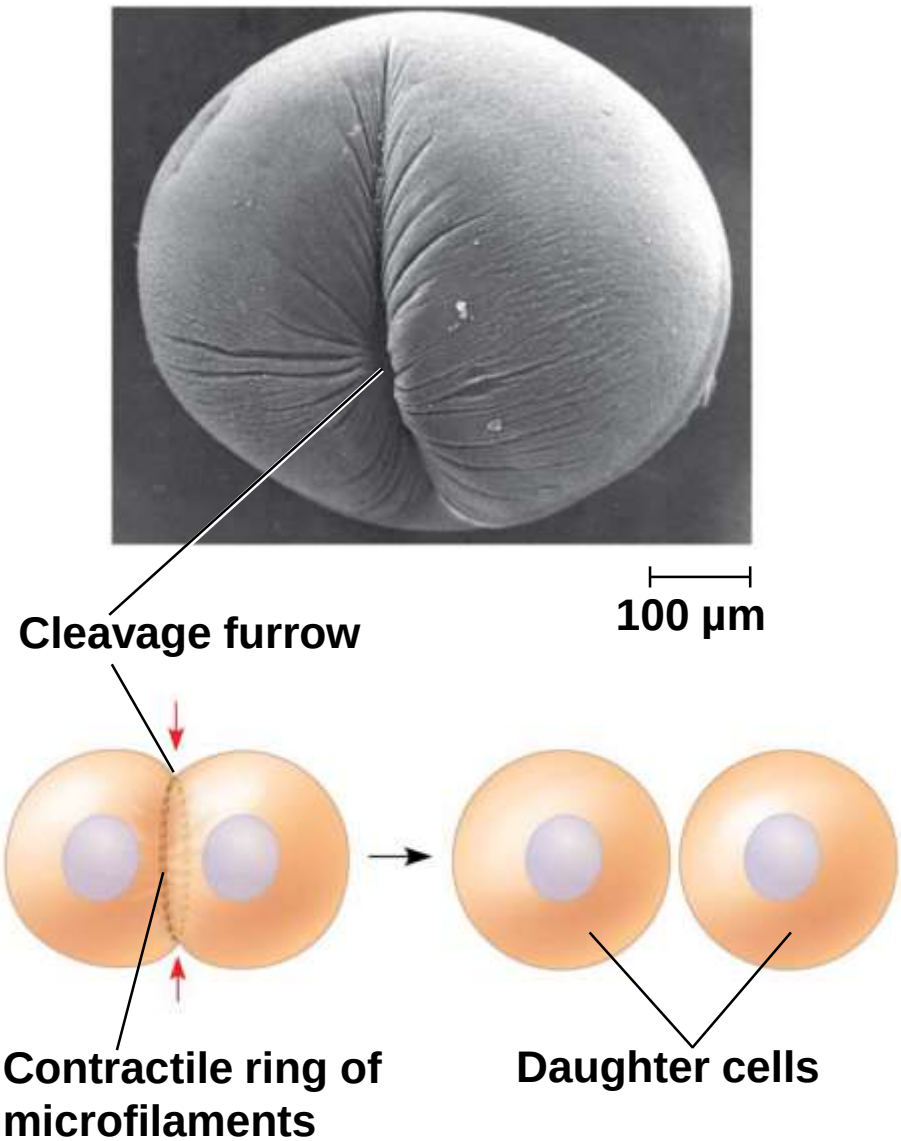
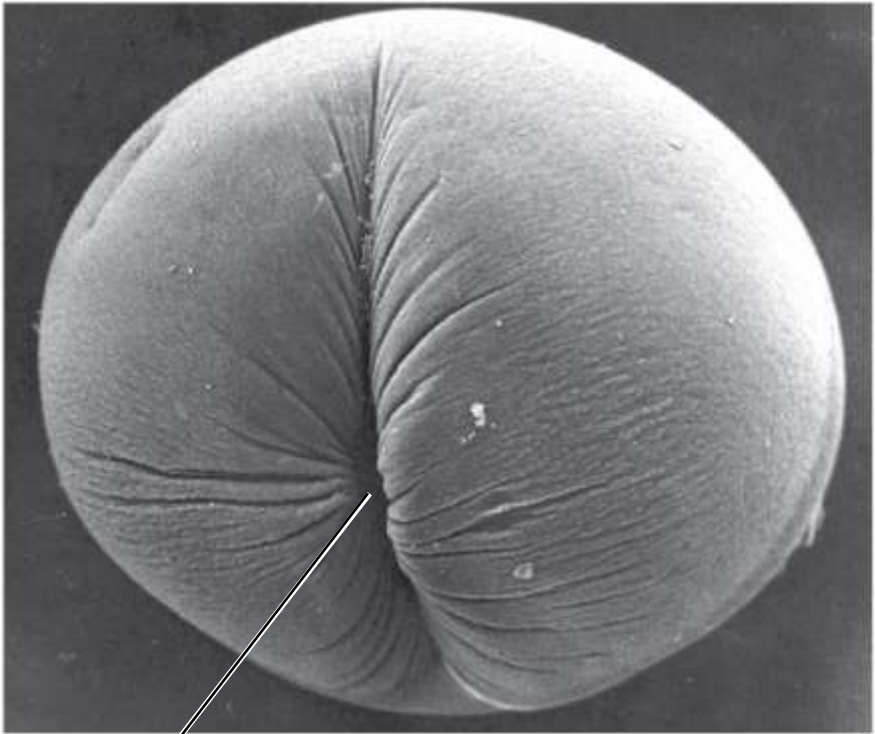


Figure 12.10aa



Cleavage furrow

100 μm

(b) Cell plate formation in a plant cell (TEM)

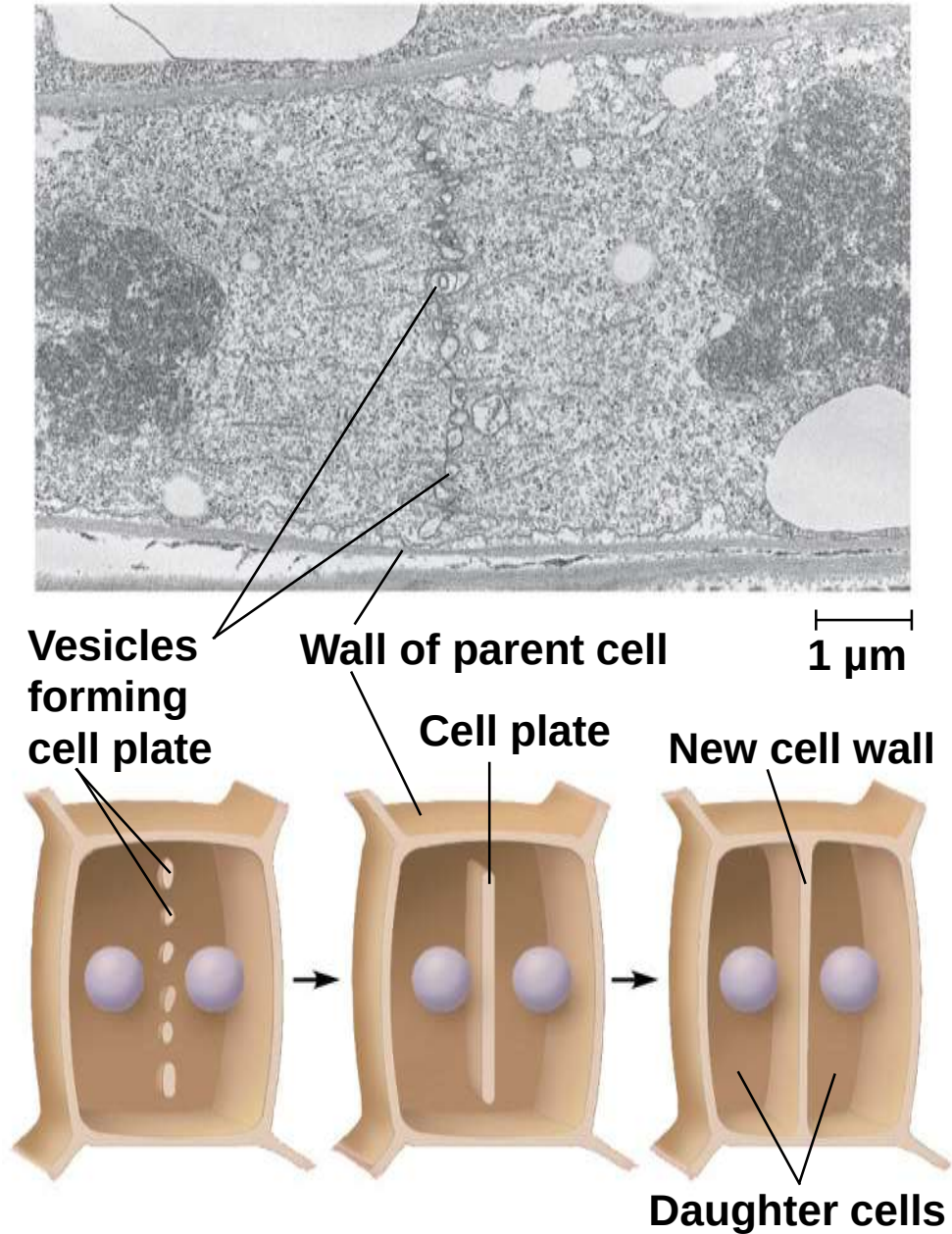
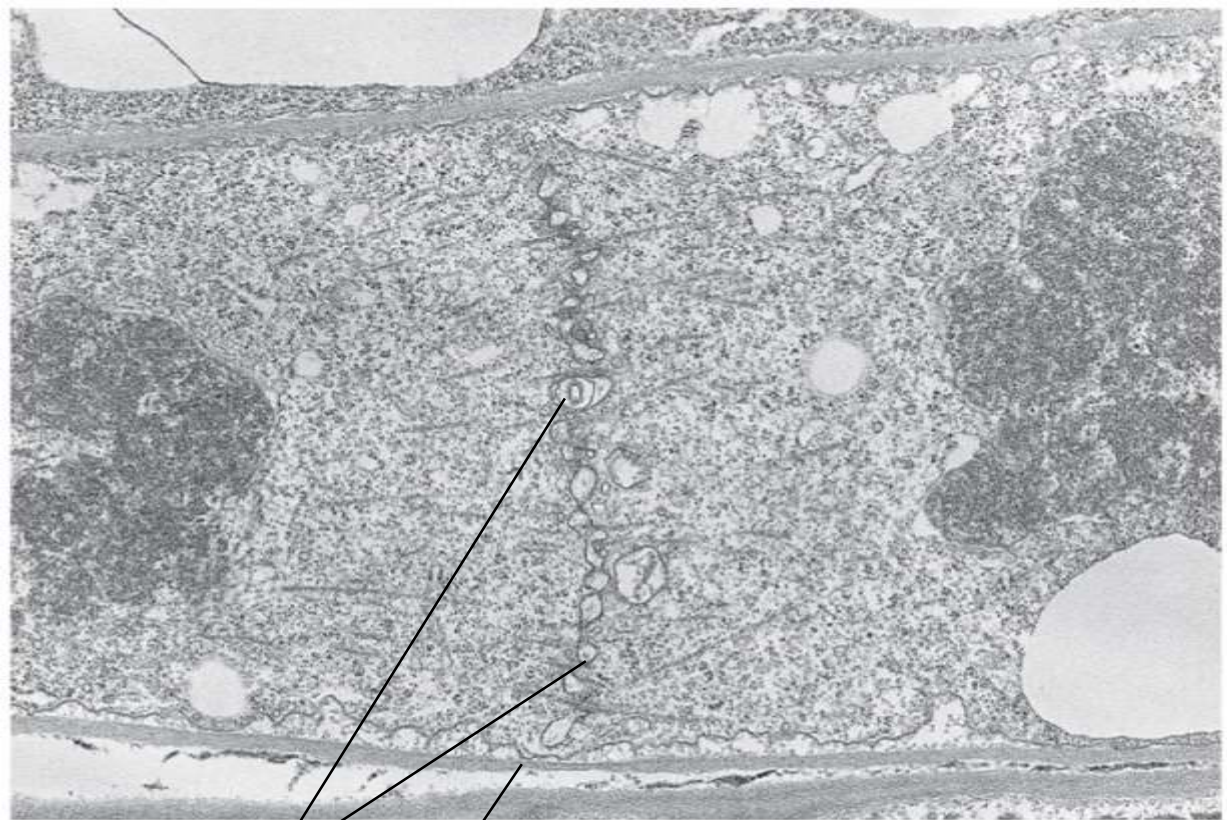


Figure 12.10ba

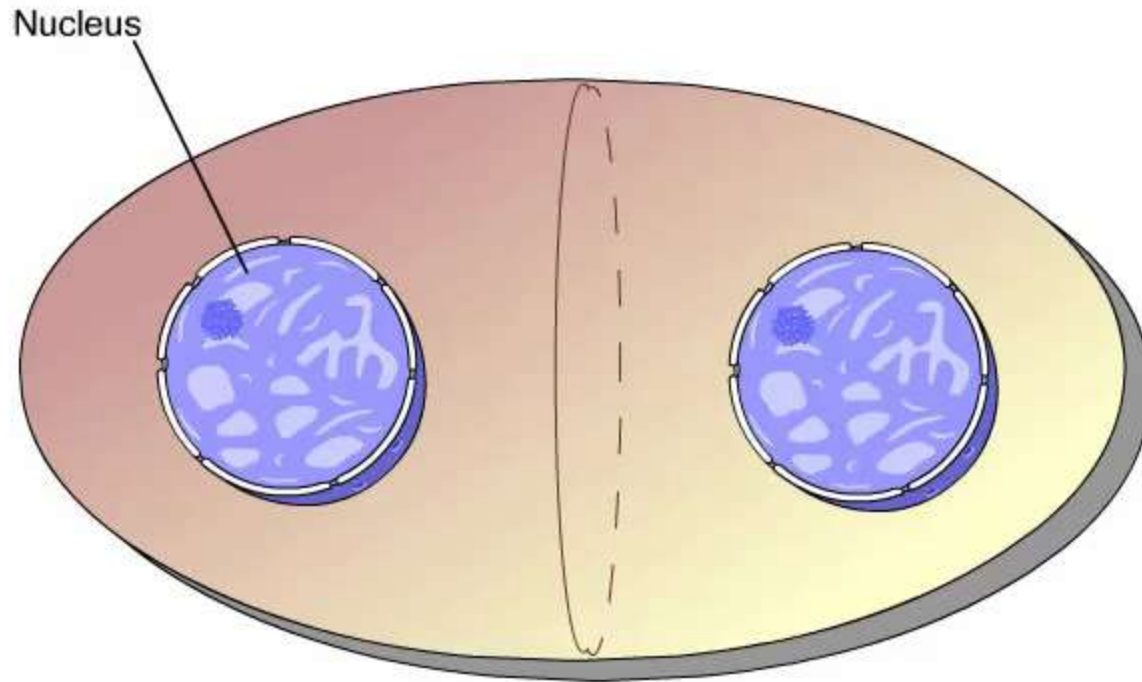


**Vesicles
forming
cell plate**

Wall of parent cell

1 μ m

Animation: Cytokinesis



CYTOKINESIS

Animal cell

Video: Myosin and Cytokinesis

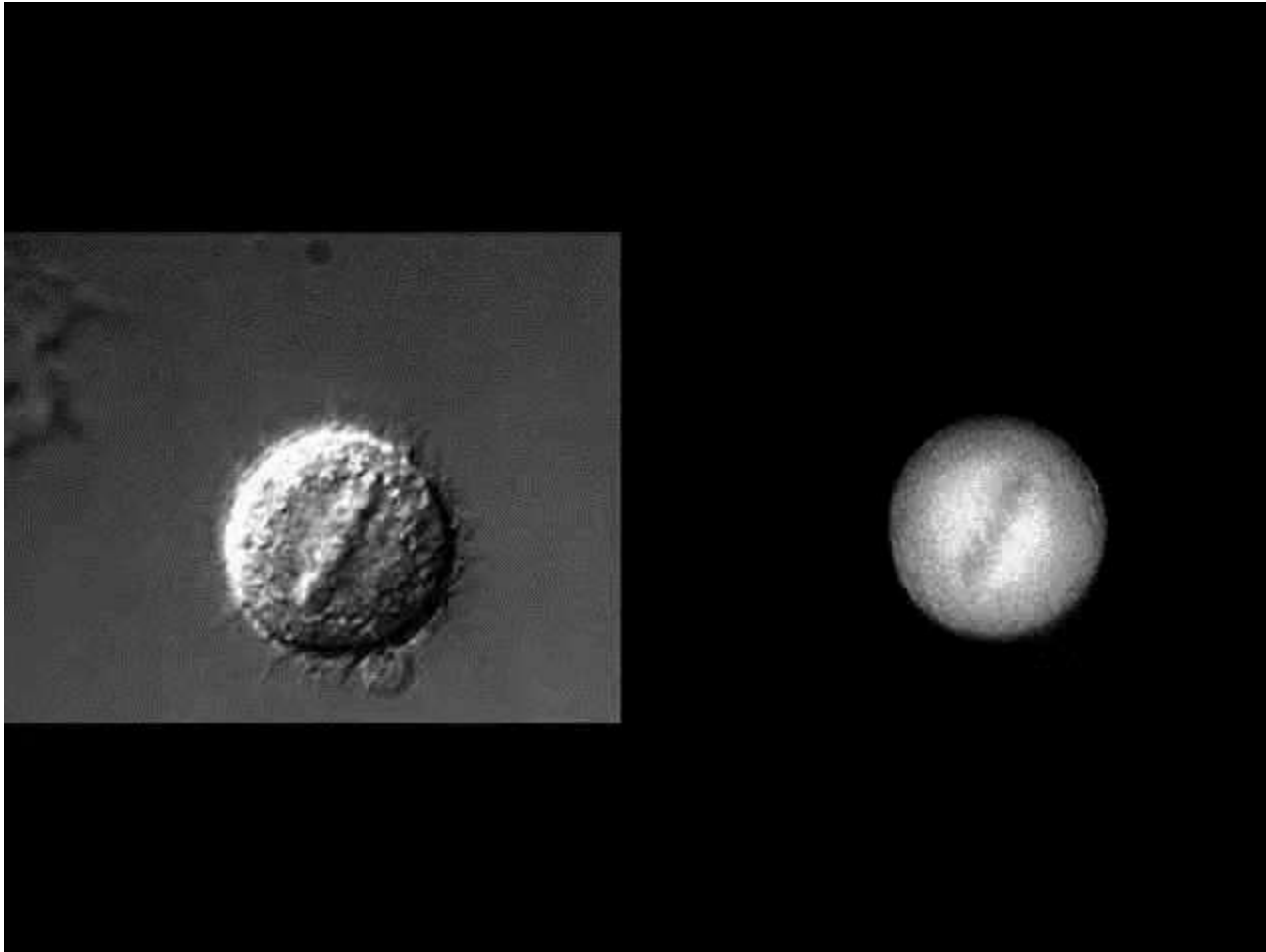
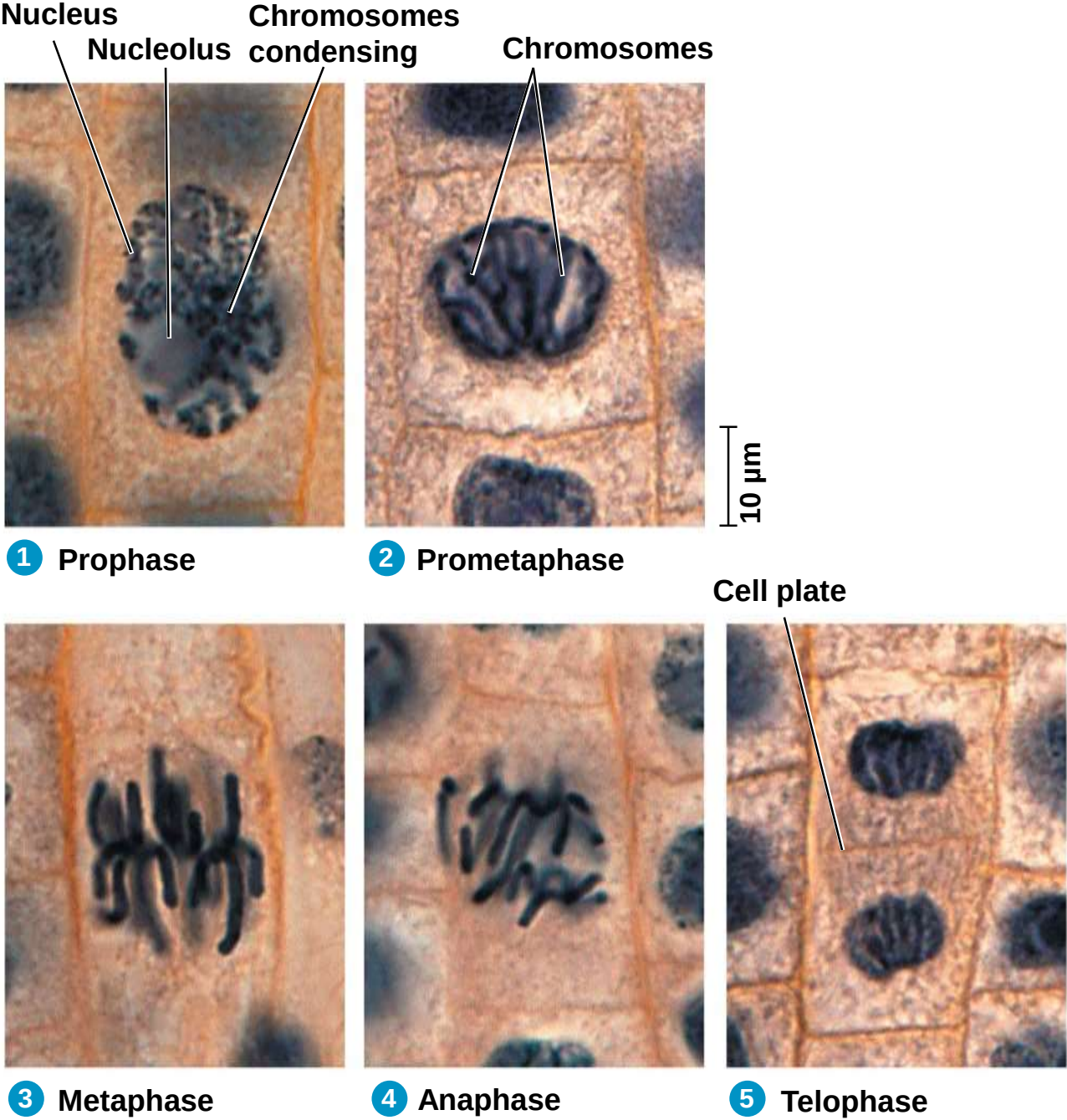


Figure 12.11



Binary Fission in Bacteria

- a) Prokaryotes (bacteria and archaea) reproduce by a type of cell division called **binary fission**
- b) In binary fission, the chromosome replicates (beginning at the **origin of replication**), and the two daughter chromosomes actively move apart
- c) The plasma membrane pinches inward, dividing the cell into two

Figure 12.12-1

1 Chromosome replication begins.

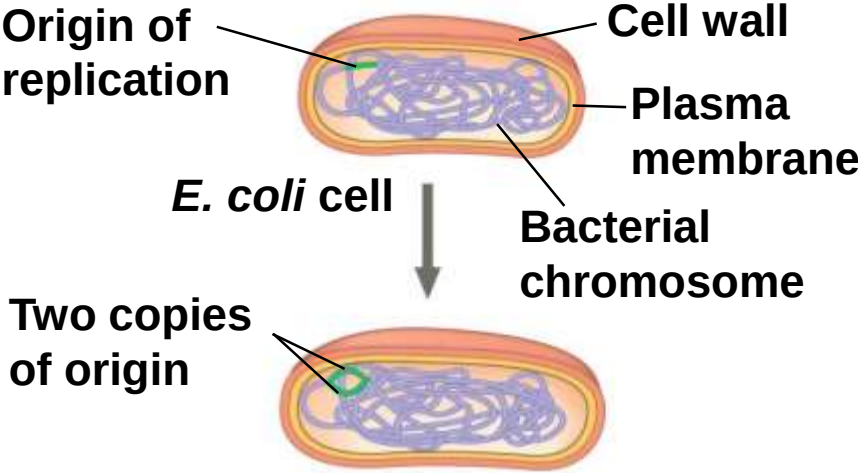


Figure 12.12-2

1 Chromosome replication begins.

2 One copy of the origin is now at each end of the cell.

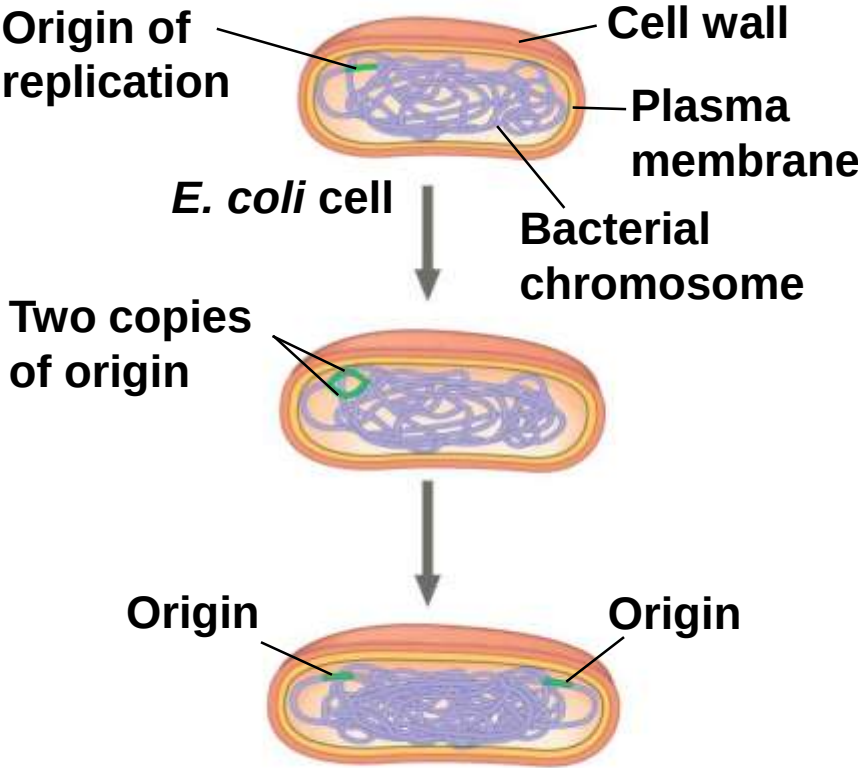


Figure 12.12-3

- 1 Chromosome replication begins.
- 2 One copy of the origin is now at each end of the cell.
- 3 Replication finishes.

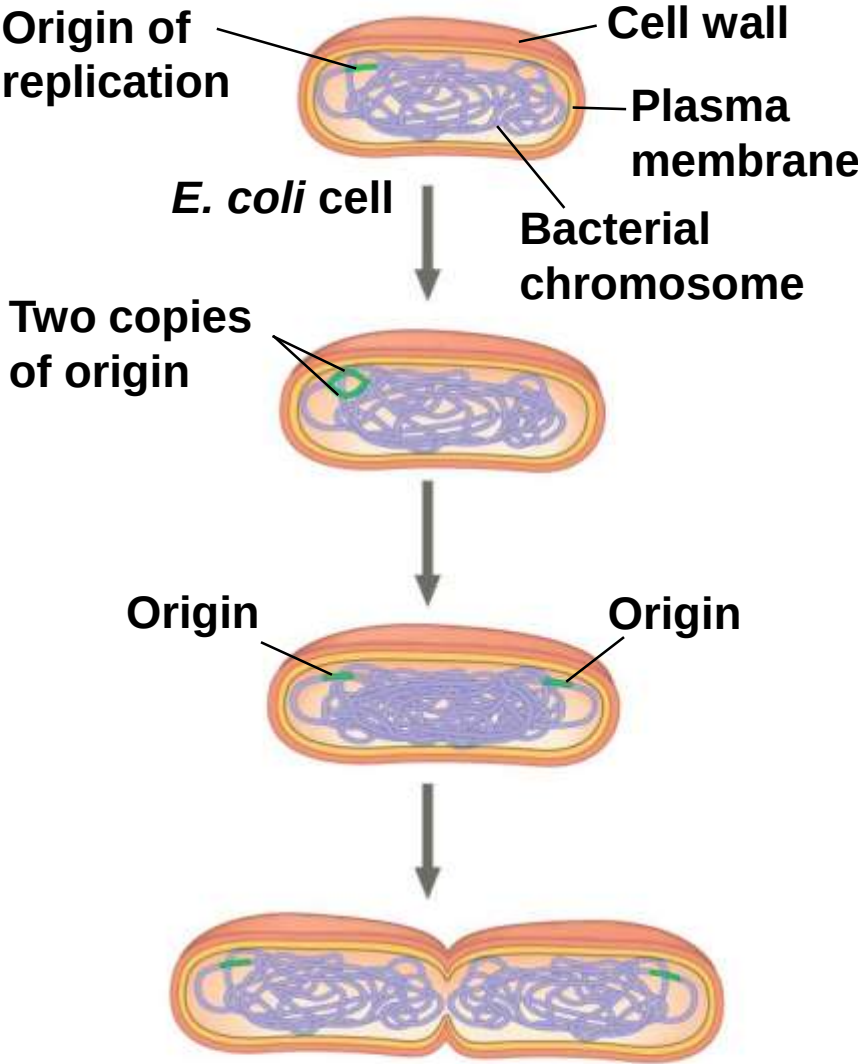
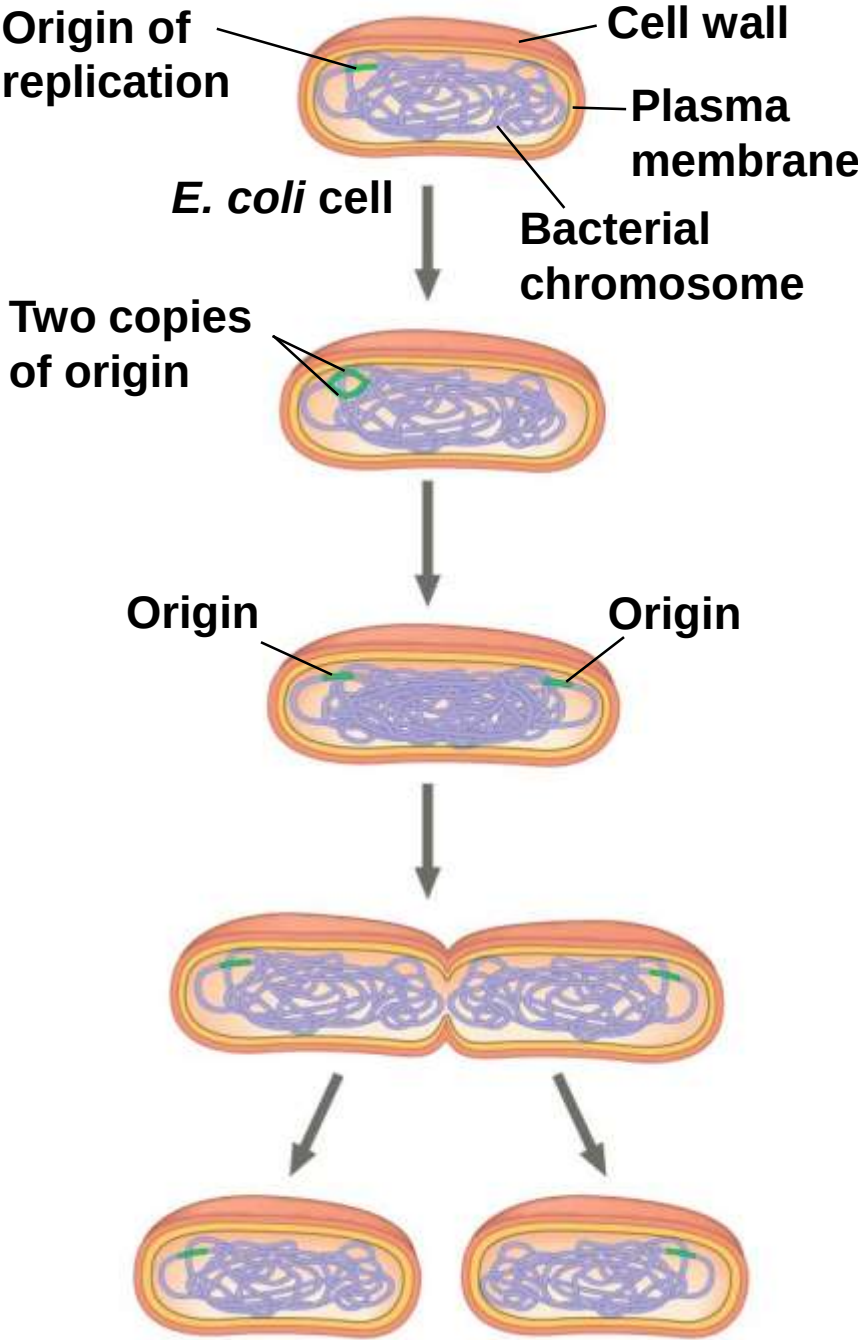


Figure 12.12-4

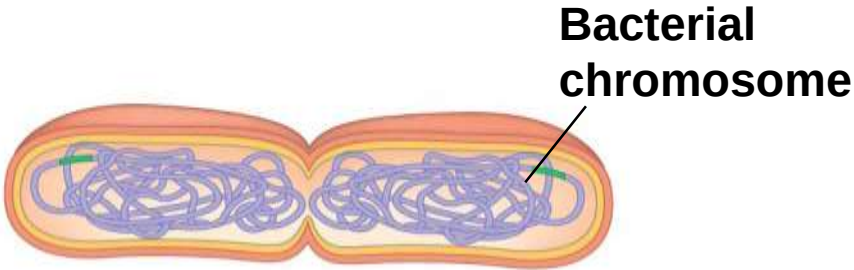
- 1 Chromosome replication begins.
- 2 One copy of the origin is now at each end of the cell.
- 3 Replication finishes.
- 4 Two daughter cells result.



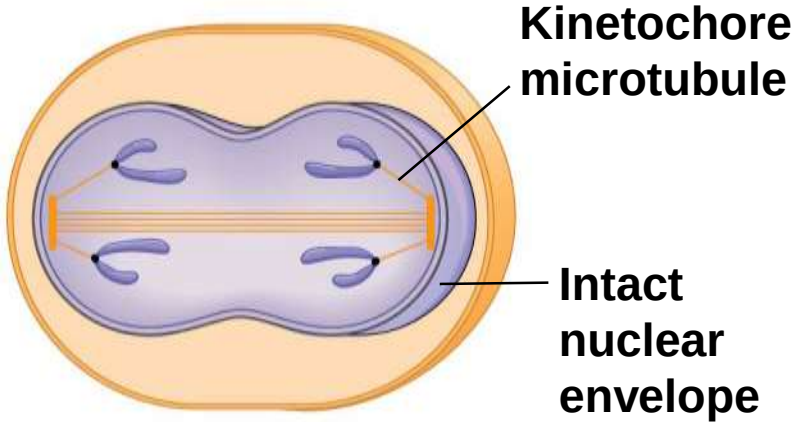
The Evolution of Mitosis

- a) Since prokaryotes evolved before eukaryotes, mitosis probably evolved from binary fission
- b) Certain protists exhibit types of cell division that seem intermediate between binary fission and mitosis

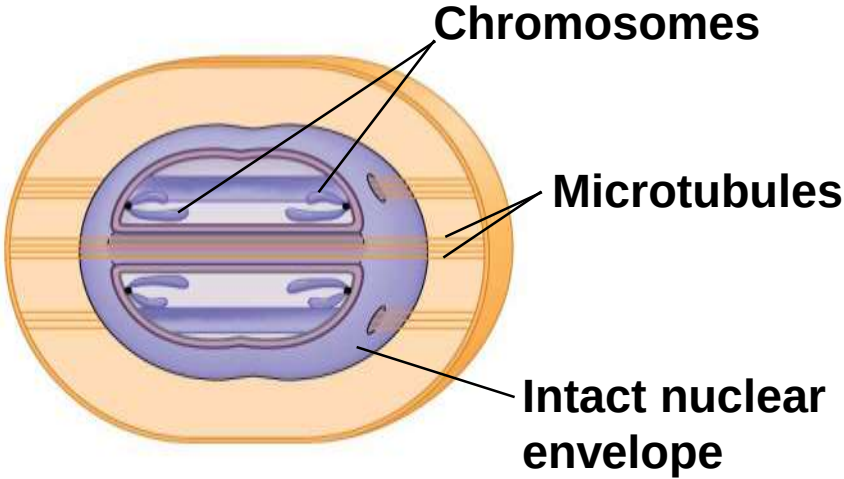
Figure 12.13



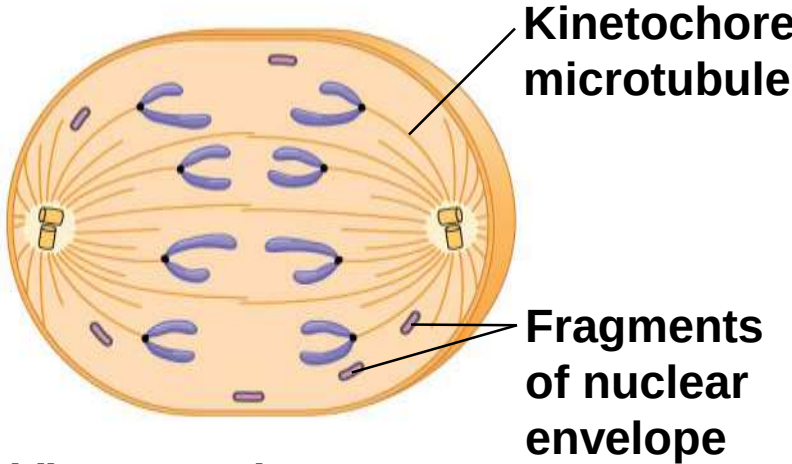
(a) Bacteria



(c) Diatoms and some yeasts



(b) Dinoflagellates



(d) Most eukaryotes

Concept 12.3: The eukaryotic cell cycle is regulated by a molecular control system

- a) The frequency of cell division varies with the type of cell
- b) These differences result from regulation at the molecular level
- c) Cancer cells manage to escape the usual controls on the cell cycle

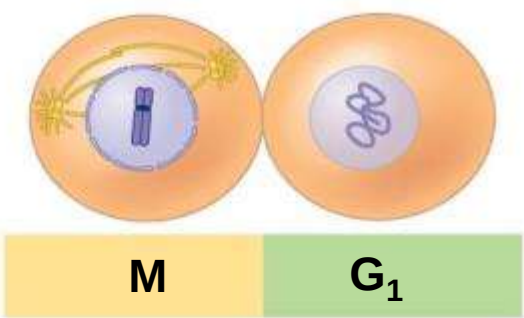
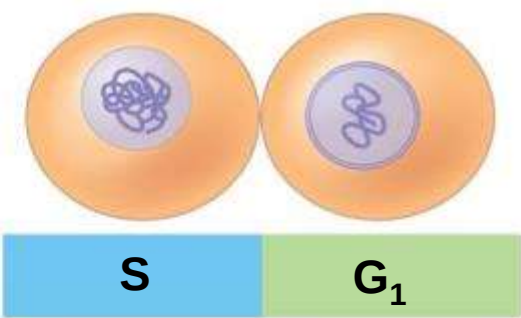
The Cell Cycle Control System

- a) The cell cycle appears to be driven by specific chemical signals present in the cytoplasm
- b) Some evidence for this hypothesis comes from experiments in which cultured mammalian cells at different phases of the cell cycle were fused to form a single cell with two nuclei

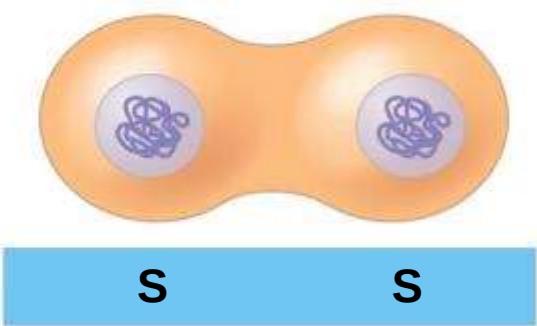
Experiment

Experiment 1

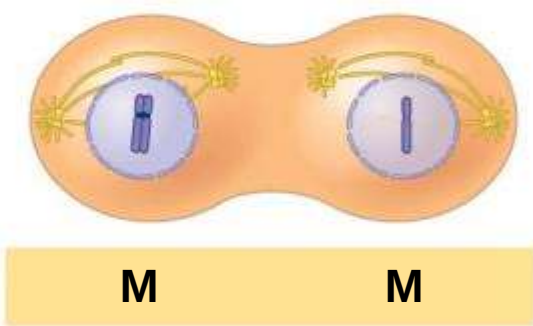
Experiment 2



Results



G₁ nucleus immediately entered S phase and DNA was synthesized.



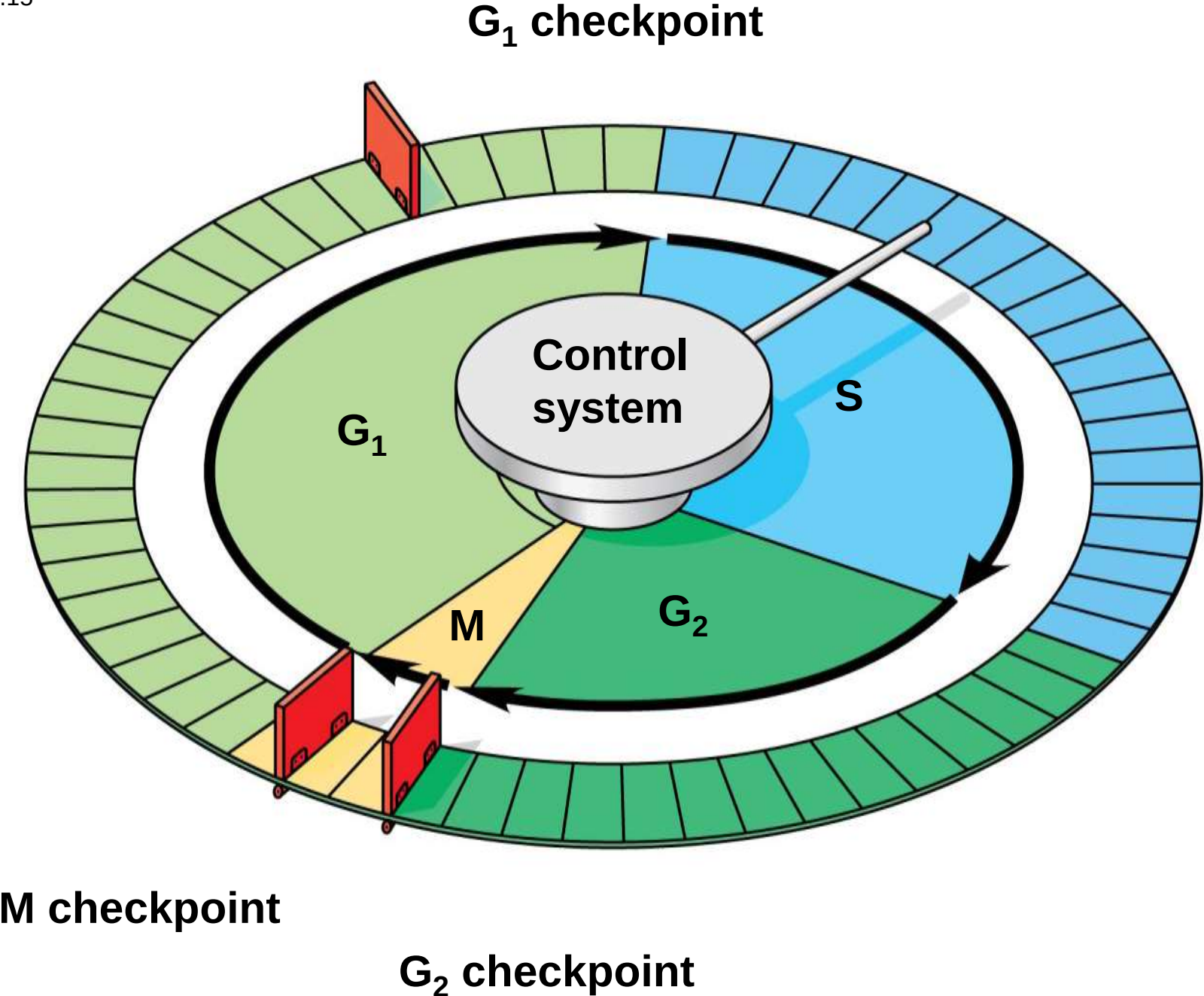
G₁ nucleus began mitosis without chromosome duplication.

Conclusion

Molecules present in the cytoplasm control the progression to S and M phases.

- a) The sequential events of the cell cycle are directed by a distinct **cell cycle control system**, which is similar to a clock
- b) The cell cycle control system is regulated by both internal and external controls
- c) The clock has specific **checkpoints** where the cell cycle stops until a go-ahead signal is received

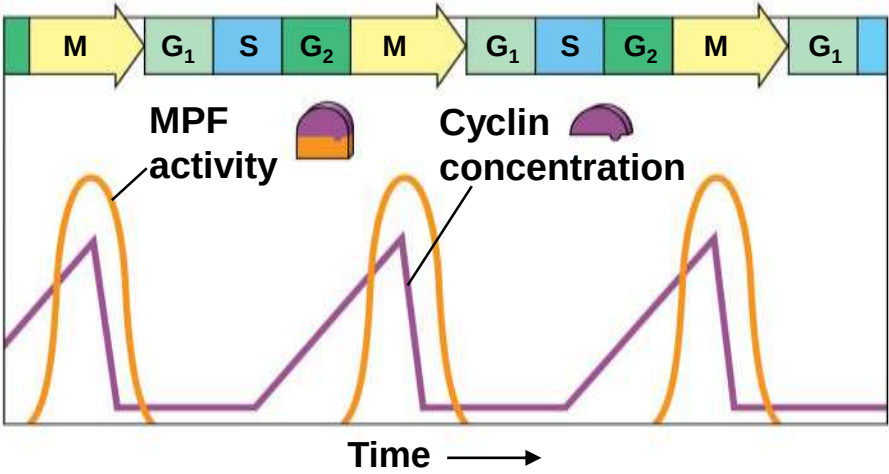
Figure 12.15



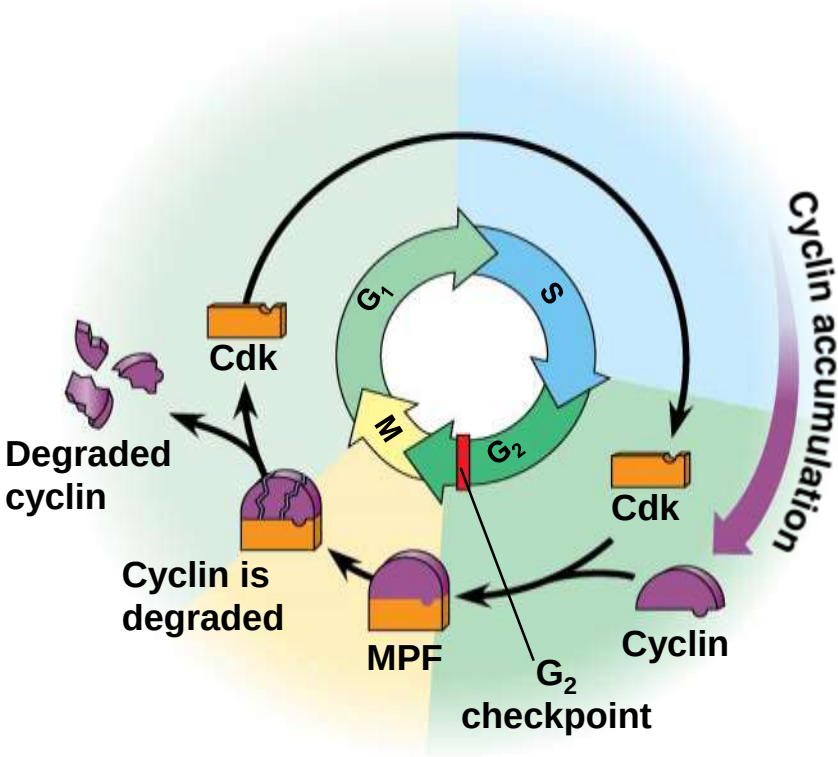
The Cell Cycle Clock: Cyclins and Cyclin-Dependent Kinases

- a) Two types of regulatory proteins are involved in cell cycle control: **cyclins** and **cyclin-dependent kinases (Cdks)**
- b) The activity of a Cdk rises and falls with changes in concentration of its cyclin partner
- c) **MPF** (maturation-promoting factor) is a cyclin-Cdk complex that triggers a cell's passage past the G_2 checkpoint into the M phase

Figure 12.16

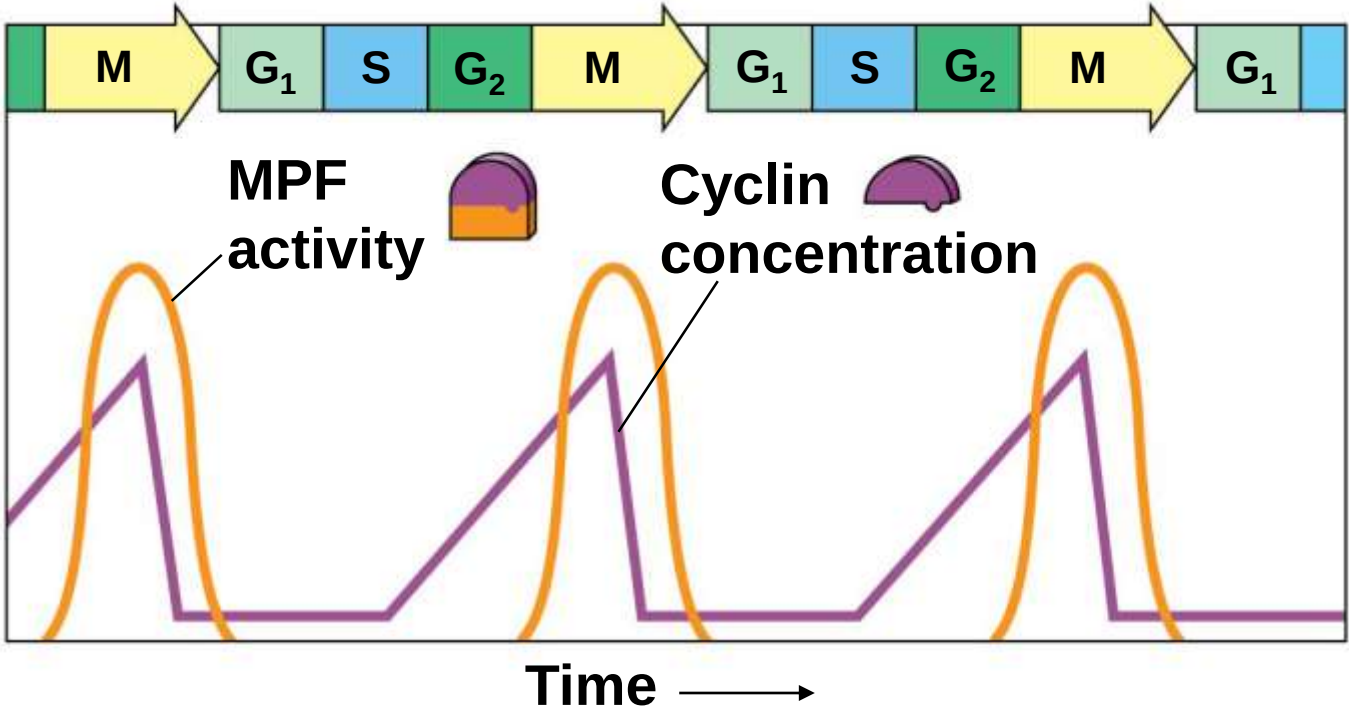


(a) Fluctuation of MPF activity and cyclin concentration during the cell cycle

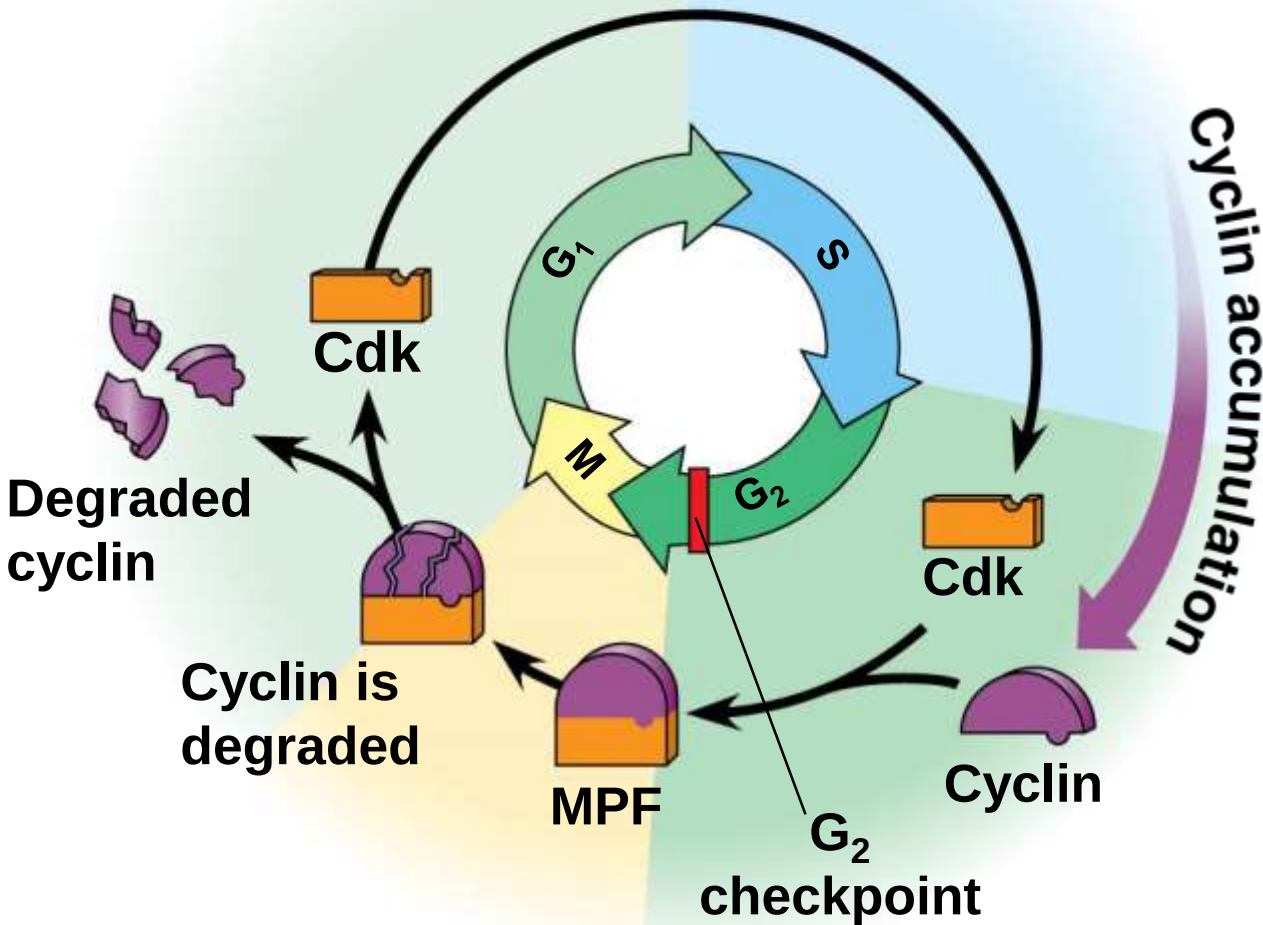


(b) Molecular mechanisms that help regulate the cell cycle

Figure 12.16a



(a) Fluctuation of MPF activity and cyclin concentration during the cell cycle



(b) Molecular mechanisms that help regulate the cell cycle

Stop and Go Signs: Internal and External Signals at the Checkpoints

- a) Many signals registered at checkpoints come from cellular surveillance mechanisms within the cell
- b) Checkpoints also register signals from outside the cell
- c) Three important checkpoints are those in G_1 , G_2 , and M phases

- a) For many cells, the G_1 checkpoint seems to be the most important
- b) If a cell receives a go-ahead signal at the G_1 checkpoint, it will usually complete the S, G_2 , and M phases and divide
- c) If the cell does not receive the go-ahead signal, it will exit the cycle, switching into a nondividing state called the **G_0 phase**

Figure 12.17

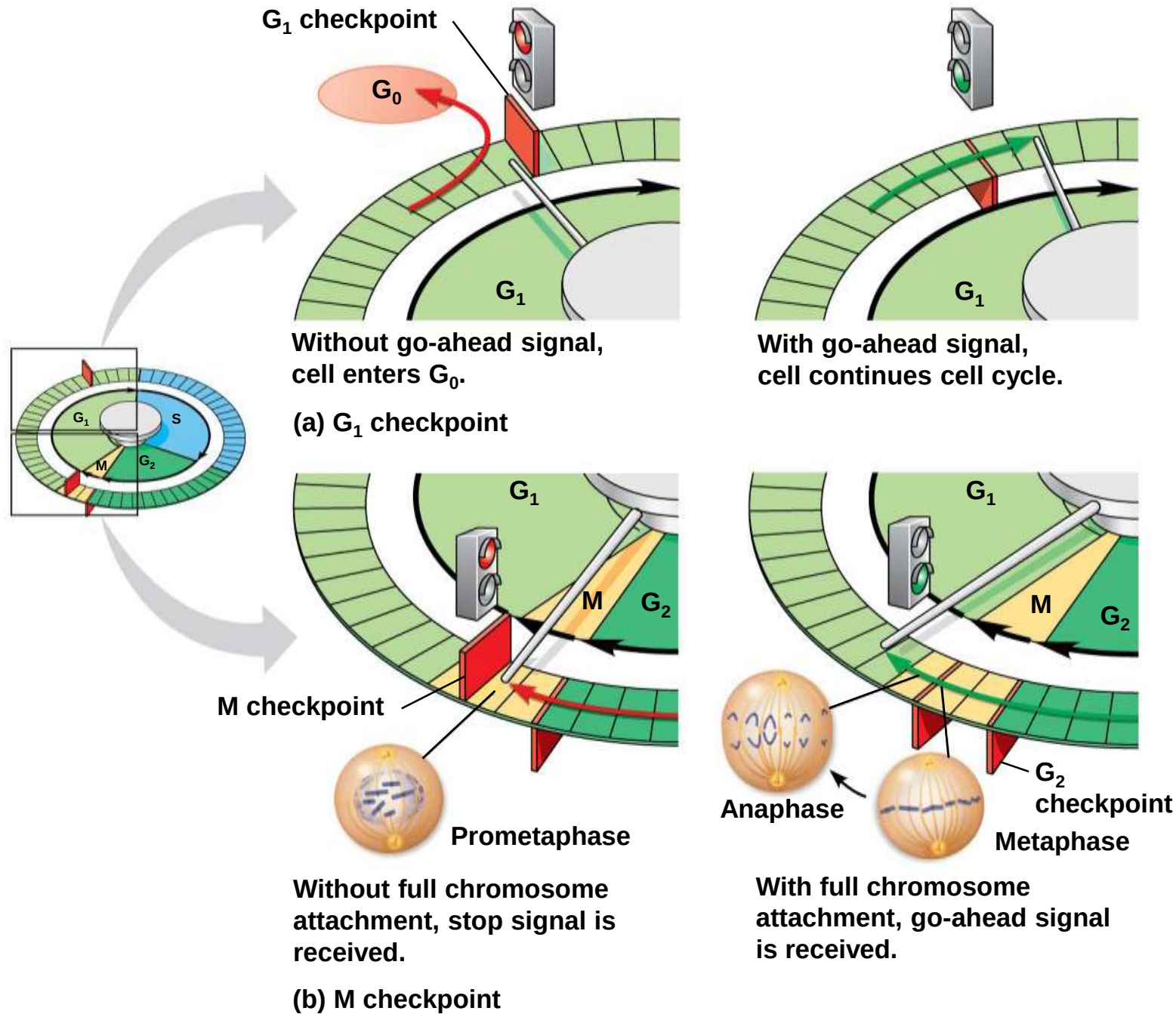
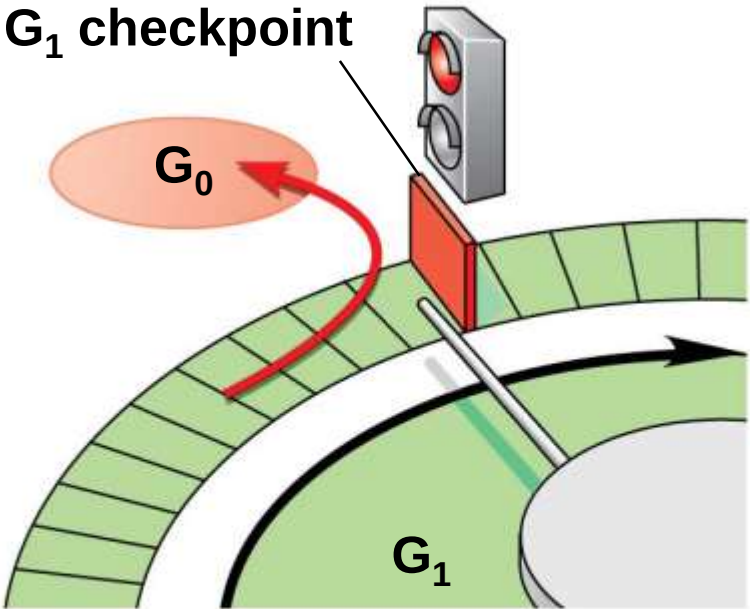
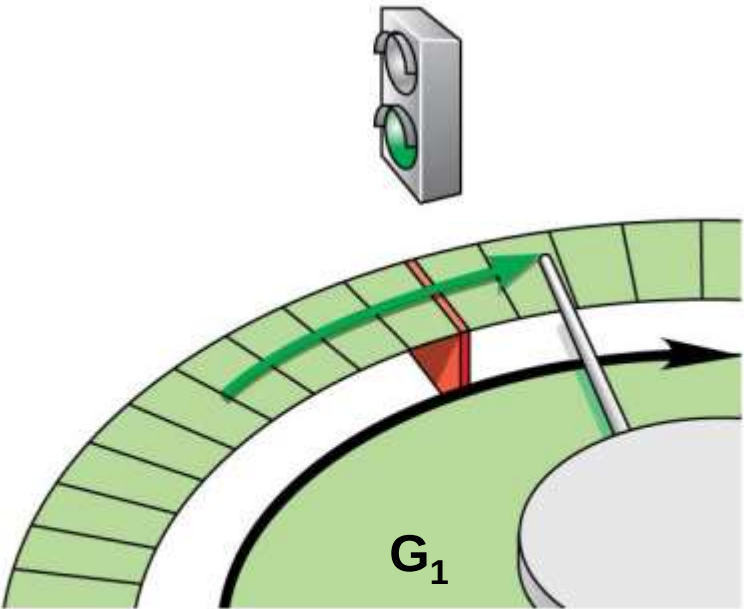


Figure 12.17a



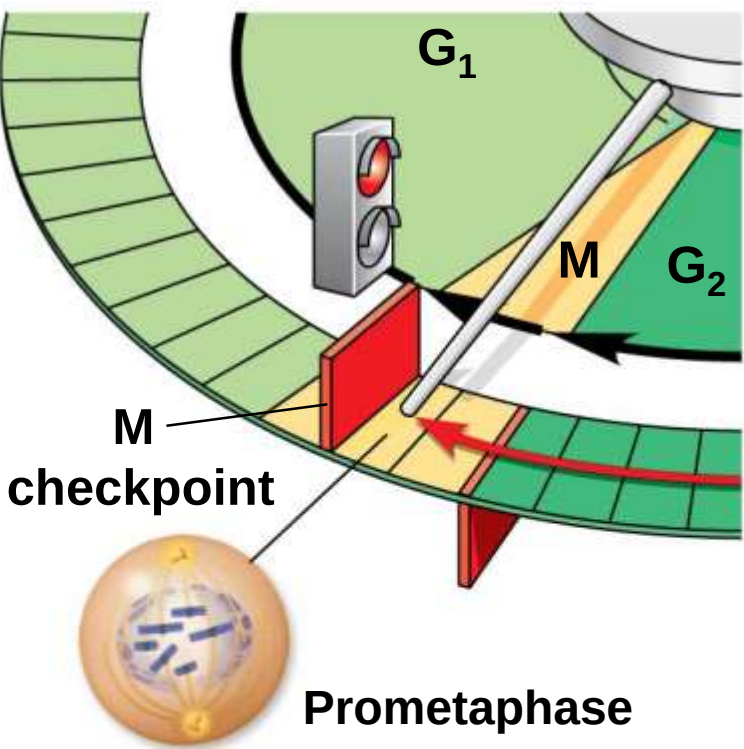
Without go-ahead signal,
cell enters G_0 .

(a) G_1 checkpoint



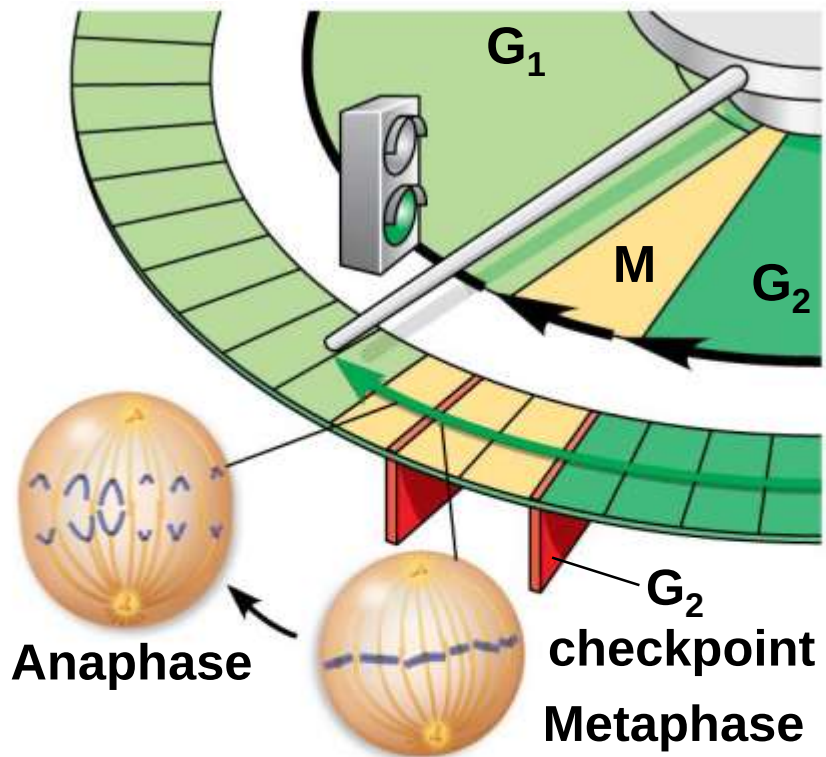
With go-ahead signal,
cell continues cell cycle.

Figure 12.17b



Without full chromosome attachment, stop signal is received.

(b) M checkpoint



With full chromosome attachment, go-ahead signal is received.

- a) An example of an internal signal is that cells will not begin anaphase until all chromosomes are properly attached to the spindle at the metaphase plate
- b) This mechanism assures that daughter cells have the correct number of chromosomes

- a) External factors that influence cell division include specific growth factors
- b) **Growth factors** are released by certain cells and stimulate other cells to divide
- c) Platelet-derived growth factor (PDGF) is made by blood cell fragments called platelets
- d) In **density-dependent inhibition**, crowded cells will stop dividing

Figure 12.18-1

- 1 A sample of human connective tissue is cut up into small pieces.

Petri dish

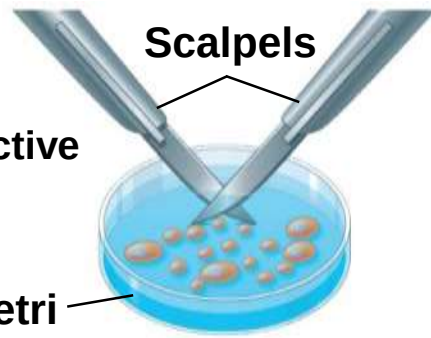
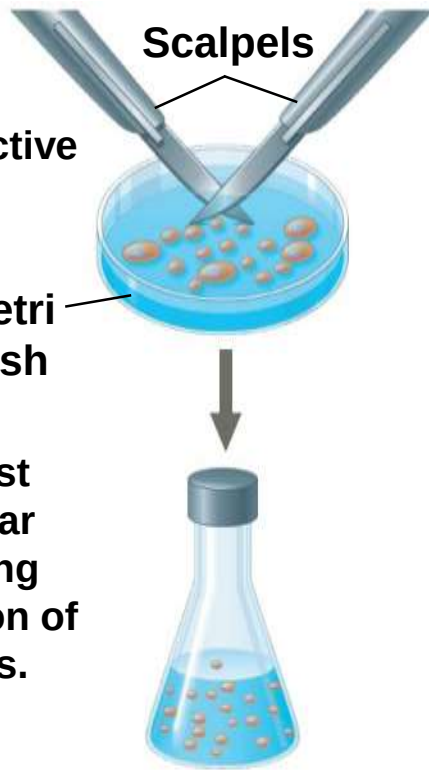


Figure 12.18-2

- 1 A sample of human connective tissue is cut up into small pieces.

Petri dish



- 2 Enzymes digest the extracellular matrix, resulting in a suspension of free fibroblasts.

Figure 12.18-3

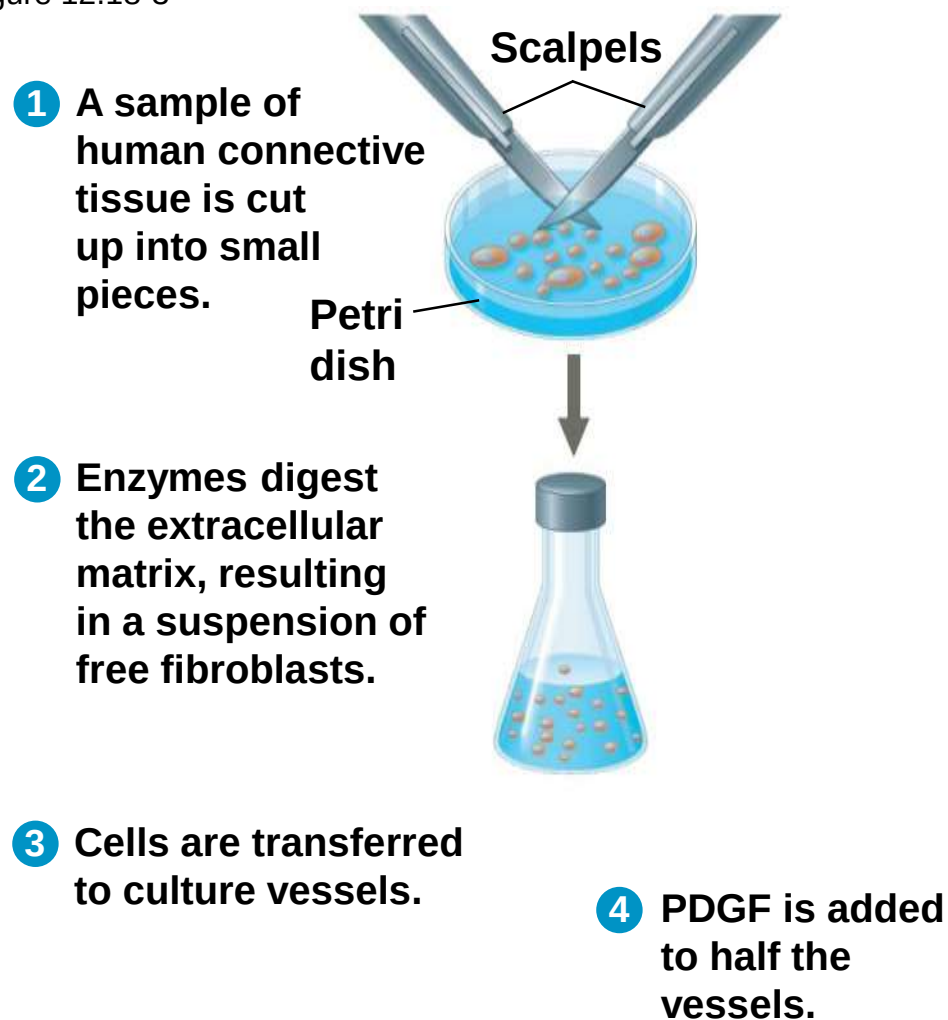
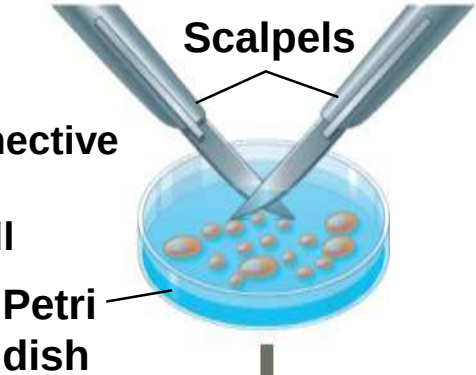


Figure 12.18-4

- 1** A sample of human connective tissue is cut up into small pieces.



- 2** Enzymes digest the extracellular matrix, resulting in a suspension of free fibroblasts.

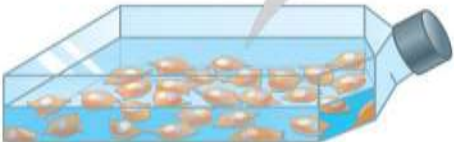


- 3** Cells are transferred to culture vessels.

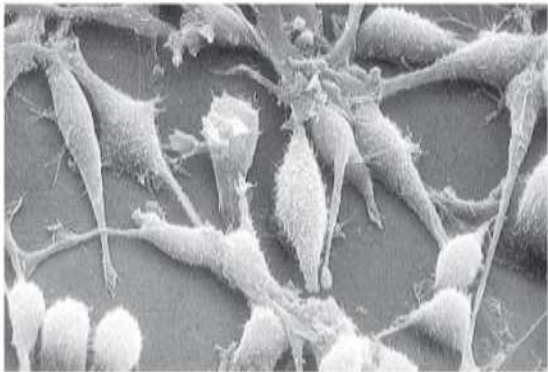
- 4** PDGF is added to half the vessels.



Without PDGF



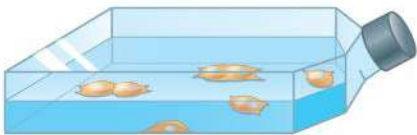
With PDGF



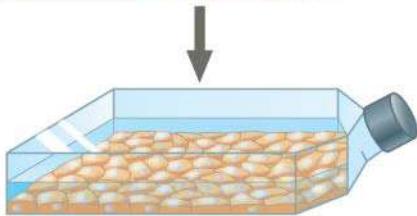
Cultured fibroblasts (SEM)

- a) Most cells also exhibit **anchorage dependence**—to divide, they must be attached to a substratum
- b) Density-dependent inhibition and anchorage dependence check the growth of cells at an optimal density
- c) Cancer cells exhibit neither type of regulation of their division

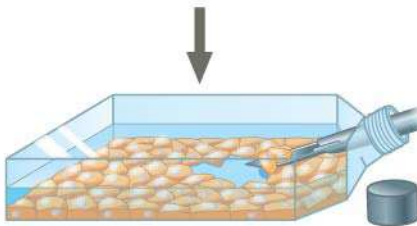
Figure 12.19



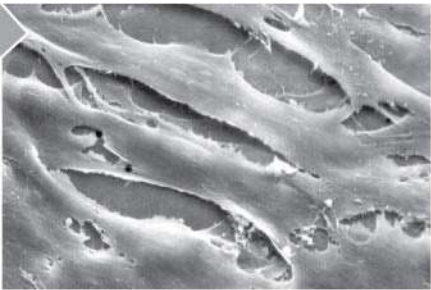
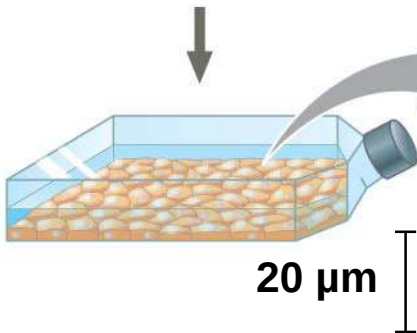
Anchorage dependence: cells require a surface for division



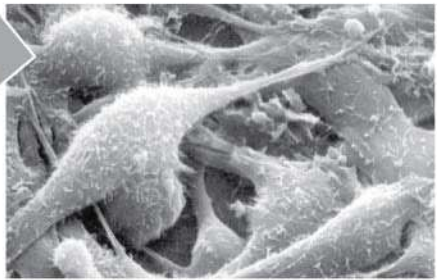
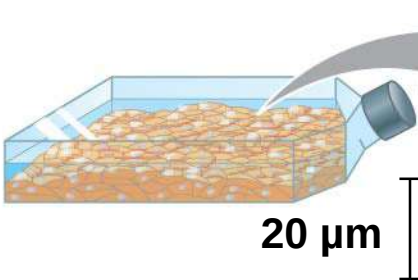
Density-dependent inhibition: cells form a single layer



Density-dependent inhibition: cells divide to fill a gap and then stop



(a) Normal mammalian cells



(b) Cancer cells

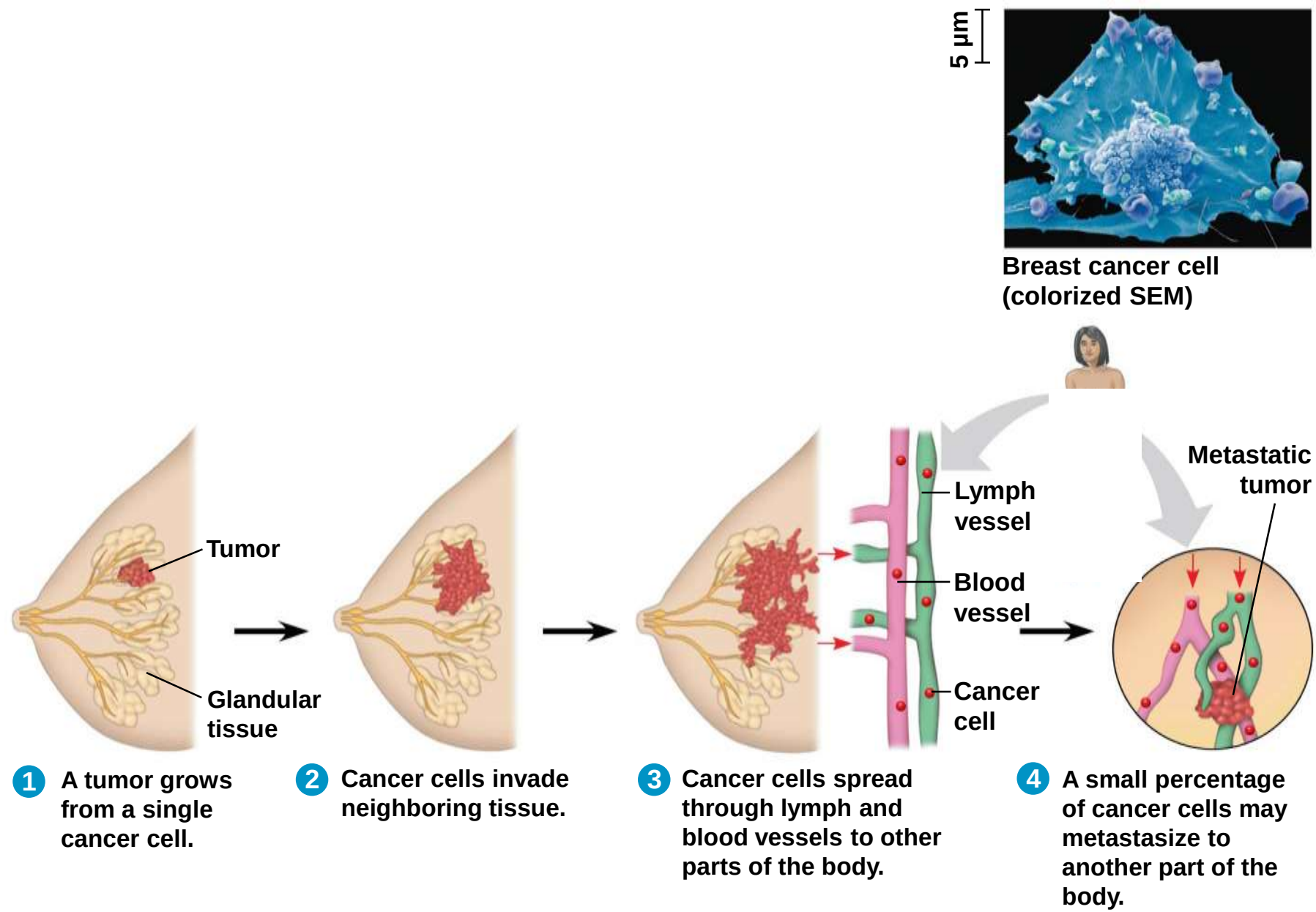
Loss of Cell Cycle Controls in Cancer Cells

- a) Cancer cells do not respond normally to the body's control mechanisms
- b) Cancer cells may not need growth factors to grow and divide
 - a) They may make their own growth factor
 - b) They may convey a growth factor's signal without the presence of the growth factor
 - c) They may have an abnormal cell cycle control system

- a) A normal cell is converted to a cancerous cell by a process called **transformation**
- b) Cancer cells that are not eliminated by the immune system form tumors, masses of abnormal cells within otherwise normal tissue
- c) If abnormal cells remain only at the original site, the lump is called a **benign tumor**

- a) **Malignant tumors** invade surrounding tissues and can **metastasize**, exporting cancer cells to other parts of the body, where they may form additional tumors
- b) Localized tumors may be treated with high-energy radiation, which damages the DNA in the cancer cells
- c) To treat metastatic cancers, chemotherapies that target the cell cycle may be used

Figure 12.20



- a) Recent advances in understanding the cell cycle and cell cycle signaling have led to advances in cancer treatment
- b) Coupled with the ability to sequence the DNA of cells in a particular tumor, treatments are becoming more “personalized”